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MASTER'S THESIS: MSC NATURAL LANGUAGE PROCESSING

Creation of a lexicon of suppletive stems for derivational morphology

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Declaration of Authorship

I, Maiwenn TELLOSA, declare that this thesis titled, “Creation of a lexicon of suppletive stems for derivational morphology” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed: Maiwenn TELLOSA

Date: August 17, 2026

A handwritten signature in black ink, appearing to read 'Maiwenn', with a long horizontal stroke extending to the right.

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Abstract

IDMC

Master

Creation of a lexicon of suppletive stems for derivational morphology

by Maiwenn TELLOSA

Stem variation is a topic on which there is no consensus in morphophonology. The concepts of *allomorphy* and *suppletion* have definitions that are quite similar, both based on the idea that the stems of two morphologically related lexemes differ from one another. This leads us to ask questions about the formal variation present in a base/derivative pair. Given the current state of definitions, it is difficult to determine whether a variation in the form of the derivative's stem constitutes allomorphy or suppletion. Our goal is to produce a resource that would help clarify these concepts. In this thesis, we present the preliminary steps toward the development of a lexicon of suppletive stems: BREF (Base de données des Radicaux supplétifs sur critère Etymologique du Français)¹.

¹Database of suppletive Stems based on Etymological criteria in French

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Chapter 1

Introduction

1.1 Host organization

This thesis reports on a work conducted during a 6-month research internship at the ATILF¹ (Analyse et Traitement Informatique de la Langue Française) (research) institute from March 02, 2026 to August 31, 2026 (26 weeks). DeepL was used as a help to translate this work into English. It was supported by the Impact INSIGHT project within the framework of the Lorraine Initiative of Excellence, a France 2030 program. This internship constitutes the author’s final semester of Natural Language Processing master’s degree in IDMC² (Institut des Sciences du Digital, Management et Cognition), Nancy. During her internship, the author was supervised by Fiammetta Namer, Stéphanie Lignon, from the *Morphologie constructionnelle* research group within the *Lexique* disciplinary axis and the *Modélisation, ressources et traitement informatique* transverse axis of the ATILF (research) institute, and Nabil Hathout, who was an external supervisor from the CLLE³ (Cognition, Langues, Langage, Ergonomie) (research) institute.

ATILF is a research institute in linguistics and is a joint research unit shared by the CNRS⁴ (Centre National de Recherche Scientifique) and the Université de Lorraine⁵. ATILF draws on strong dual expertise in linguistics and computer science, enabling it to make numerous online platforms and tools available to the public, including several dictionaries and other reference resources and tools. There are 135 members, 6 research areas and 4 research support services, including the documentation center Michel Dinet specialized in linguistics and language didactics. The objectives of the *Morphologie constructionnelle* research group are to build morphological resources and to conduct morphological analysis.

1.2 Word formation and phonological alternations

We might think that the question of derivation in morphology is now well settled, since it relies on a simple correlation between structure and meaning. In fact, let’s take the case of the suffix *-ique* in French. When it combines with a nominal base X, it forms an adjective *Xique*, whose morphologically complex meaning can be paraphrased as ‘of X, related to X’. With the pair CUBE / CUBIQUE, we can indeed analyze CUBIQUE as CUBE + -IQUE, and its meaning ‘qui a la forme d’un cube, relatif au cube’ derives directly from that of its base. However, things get more complicated

¹See <https://www.atilf.fr/>

²See <https://idmc.univ-lorraine.fr/>

³See <https://clle.univ-tlse2.fr/>

⁴See <https://www.cnrs.fr/fr>

⁵See <https://www.univ-lorraine.fr/>

when we look at other pairs based on the same X/Xique pattern. What, for example, should we make of the pairs NÉVROSE/NÉVROTIQUE, ENZYME/ENZYMATIQUE, STYLE/STYLISTIQUE, FAIM/FAMÉLIQUE, EAU/AQUATIQUE, CHEVAL/HIPPIQUE or NOM/ONOMASTIQUE? In each of these cases, the stem of the base noun is more or less altered, or even completely absent, in the *-ique* derivative: the formal link that clearly connected CUBE and CUBIQUE (/kyb/ > /kybik/) gradually weakens (/nevroz/ > /nevrotik/, /fɛ/ > /famelik/), until it disappears completely in pairs such as CHEVAL/HIPPIQUE, where /ʃəval/ and /ip/ no longer share any phonemes. This raises several questions. Where does the *-at-* sequence that we see in ENZYMATIQUE come from, or the *-ist-* sequence in STYLISTIQUE? This type of element cannot be predicted based solely on the base (ENZYME, STYLE). Are they part of the base, a linking element, or perhaps the trace of an ancient etymon? More broadly, how can a speaker, or cannot, establish a connection between EAU and AQUATIQUE, or between NOM and ONOMASTIQUE? Are these relationships of the same nature as the one linking CUBE and CUBIQUE, or is it something else entirely? Are we even still within the scope of morphology?

These questions tie into a broader debate in word formation, between two concepts used to describe stem variations that have yet to reach a consensus in the literature: allomorphy and suppletion. Allomorphy refers to variants of a form (e.g., FLEURISTE and FLORAL are two derivatives of FLEUR, with the latter exhibiting only a minimal stem variation), while suppletion refers to a complete change of stem (e.g., CHEVALIN and HIPPIQUE are two derivatives of CHEVAL, with the second having a stem that is completely different from the base). It is precisely around stems that exhibit high formal variation that the central issue of this thesis is developed.

1.3 Issue addressed in this thesis: developing “BREF”

In the literature, the concepts of allomorphy and suppletion are generally placed on a continuum, and numerous criteria have been proposed in an attempt to distinguish between them (see Section 2.1.5 for a detailed presentation). Traditionally, the criterion of formal distance is used to make this distinction (a high distance corresponds to a suppletive phenomenon, and conversely, a low distance corresponds to an allomorphic phenomenon). Instead, we propose to approach these phenomena from a diachronic perspective, which will allow us to distinguish suppletion from allomorphy, and then we will combine this with measurable criteria. This means that one of the stems of a lexeme may be considered suppletive because it is etymologically related (e.g., /flor/ in FLORAL comes from the Latin *flos, floris* ‘flower’), but according to other criteria, such as the criterion of formal distance, it may also be considered allomorphic.

To conduct an analysis of these phenomena of stem variation, we need to have access to appropriate resources. However, at present, there is a lack of resources that describe morphophonological variations in a valid, consistent, and systematic manner. The overall objective is therefore twofold: 1) to address this gap by developing appropriate lexical resources, and then 2) to use these resources to contribute new insights to the debate on allomorphy *versus* suppletion.

This thesis focuses on the first objective, more specifically on the construction of BREF (Base de données des Radicaux supplétifs sur critère Étymologique du Français), which will provide a documented inventory of the suppletive stems of

each lexeme. For example, provide stems such as *hipp-* or *aquat-*, whose form is directly inherited or borrowed from an ancient etymon (*hipp-* from the Ancient Greek *hippos* ‘horse’).

1.4 Context

Démonette⁶ serves as the starting point for this research project: this French derivational database brings together harmonized lexical resources and contains 222,118 lexeme pairs organized into derivational families (Namer et al., 2023). Structured as co-indexed tables (lexemes and relations), it describes the morphological links between lexemes within the same family. In line with this, GLeMM⁷ (Gros Lexique Morphologique Multilingue) is a very large-scale resource whose entries consist of pairs of morphophonologically related lexemes, enriched with morphological and semantic annotations for a portion of the data (Hathout et al., 2025). However, the fully automated construction of this database relies on relationships of formal analogy between words within the same derivational family, which effectively excludes suppletive relationships, such as *JEU* / *LUDIQUE*. The relevance of the structure and format of a third resource, Phononette, has already been demonstrated with a pilot version (Namer, Hathout, and Lignon, 2017), followed by a version 2 (Namer, Lignon, and Hathout, 2025) utilizing 90,845 entries from Démonette that were annotated morphophonologically. However, this resource remains limited by the lack of complete phonological coverage of the Démonette entries, a major obstacle for quantitative analyses, and needs to be enriched with suppletive data. Since then, with GLeMM-3⁸, the phonemic transcriptions have been completely recalculated, providing both standardized and consistent IPA (International Phonetic Alphabet) transcriptions and the derivational relationships between lexemes, which makes it possible to envisage the creation of a final version: Phononette-3. It is against this backdrop that the work of this thesis takes place: the development of BREF. In fact, the creation of Phononette-3 begins with an initial step of extracting all the suppletive relationships present in the French lexicon by drawing on scattered lexical and derivational resources. Figure 1.1 illustrates the key steps of this global project.

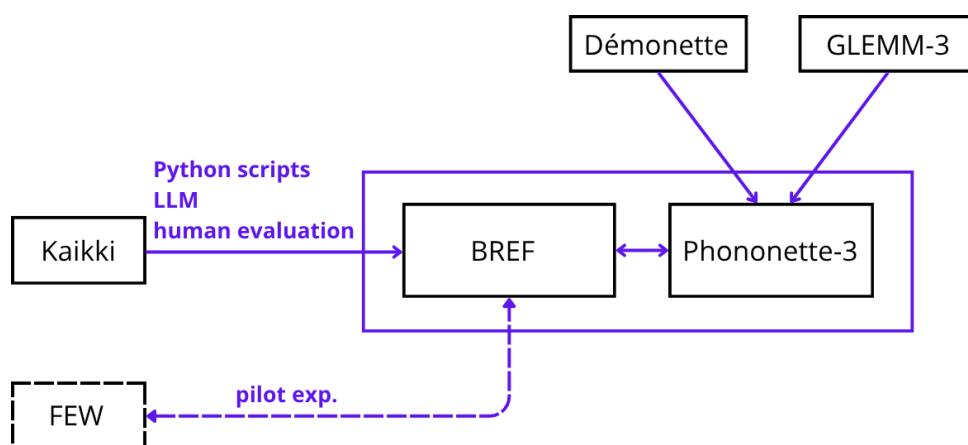


FIGURE 1.1: The key steps of the global project

⁶See https://demonette.fr/demonext/vues/front_page.php

⁷See <https://src.koda.cnrs.fr/nabil.hathout.1/glemm>

⁸See https://src.koda.cnrs.fr/nabil.hathout.1/glemm_3

1.5 Outline

Chapter 2 presents the assumptions necessary for understanding this work, particularly with regard to allomorphy and suppletion. Chapter 3 presents the general methodology adopted to build this lexicon of suppletive stems (Section 3.1), based on the extraction of etymological data using LLMs (Large Language Models) (Section 3.1.2) from the Kaikki resource (Section 3.1.1). Before building the future lexicon that will list all the suppletive stems, we first use an intermediate dataset that contains all relevant information regarding the stems (Section 3.2) and for which careful consideration was given to determine: 1) the context required for the LLM to find all the necessary information (Section 3.2.1), 2) the specific information the LLM needed to find (Section 3.2.2), and 3) the prompt that needed to be crafted for the LLM to respond effectively (Section 3.2.3). A gold standard dataset is created to evaluate the quality of the intermediate dataset (Section 3.3). Building on this, Chapter 4 presents the initial results and their human (Section 4.1) and automated (Section 4.2) evaluations. Based on the conclusions drawn from the first results, Chapter 5 presents how to improve performance by refining the context provided to the LLM (Section 5.1) and by testing different types of models (Section 5.2). Finally, Chapter 6 discusses the limitations (Section 6.1), summarizes the various results obtained during this work (Section 6.2) and future directions (Section 6.3).

All the data and scripts are available on the following GitHub repository: <https://github.com/maiWenn217/Phononette>.

Chapter 2

Background

2.1 Morphological presuppositions

2.1.1 What is morphology?

Morphology is the branch of linguistics that studies the form and meaning of words. It helps us understand and explain how the lexicon¹ is organized and constructed. In a lexeme-based approach (Aronoff, 1976; Aronoff, 1994; Fradin, 2003; Roché, 2011), in which this work is situated, the “word” constitutes the fundamental linguistic unit of morphological analysis. In this thesis, we will adopt the more rigorous terminology proposed by Matthews (1991), who, rather than using the word “word” from everyday vocabulary, prefers to use the terms “lexeme” and “wordform”.

Lexeme: lexical unit abstracted from its inflectional variations and belonging to a major lexical category (noun, verb, adjective) (e.g., CHEVAL).

Wordform: form used in speech realized by a prosodically autonomous unit belonging to a major lexical category (noun, verb, adjective) and bearing all the markers required by the syntactic and/or enunciative context (e.g., *cheval* and *chevaux*).

In other words, the lexeme is the canonical form that represents all the forms a word can take in a sentence: the wordforms. In accordance with standard conventions in morphology (Matthews, 1991), wordforms are written in italics, while lexemes are written in small caps. Other arbitrary conventions have been established in this thesis. The graphical representation of the sequence remaining after the removal of an affix (prefix or suffix), which we will call the stem, is set in a sans-serif font. For example, for CHANTEUR, what remains after removing the suffix *-eur* is the sequence chant-. The etymons² are set in a teletype font. For example, ÉQUESTRE comes from the Latin etymon equus.

Consequently, morphology is simply the study of how the form and meaning of **lexemes** and **wordforms** structurally complex vary jointly in a regular manner, such that their meaning can be (partially) inferred from their base.

¹The structured set of words that speakers are able to store (mentally) and use throughout their live (which is constantly evolving).

²An attested or reconstructed word that serves as the basis for the etymology of a given term. The Latin “amare” is the etymon of the French “aimer”.

2.1.2 Inflection and derivation

Morphology is divided into two major subfields: inflectional morphology and word formation.

Inflectional morphology: one of the two major subfields of morphology that focuses on the relationships among the inflected forms of a lexeme.

For example, it examines the relationships between *danse*, *dansons*, *danserait*, and *dansent*, which are all different forms (different conjugations) of the same verb, *DANSER*, meaning ‘to perform body movements in rhythm with music’.

Word formation: one of the two major subfields of morphology that focuses on the regular relationships between form and meaning in morphologically related lexemes.

For example, it is interested in the relationship between *AGRÉABLE* and *DÉSAGRÉABLE*, which are morphologically related in terms of form (*AGRÉABLE* is found in *DÉSAGRÉABLE*) and in terms of meaning, since *DÉSAGRÉABLE* means ‘which/who is not agreeable’.

While inflectional morphology regards single lexemes, producing only different word-forms dictated by syntax, word formation contributes to lexical enrichment by creating new lexemes. These new lexemes can be formed through derivation or compounding.

Derivation: case of lexeme formation in which morphologically complex lexemes are formed from a single base (e.g., *AGRÉABLE* → *DÉSAGRÉABLE*).

Compounding: case of lexeme formation in which morphologically complex lexemes are formed from, at least, two bases (e.g., *LAVER* + *VAISSELLE* → *LAVE-VAISSELLE*).

Base: lexeme that is formally and semantically related to a morphologically more complex lexeme (e.g., *AGRÉABLE* is the base of the derivative *DÉSAGRÉABLE*).

In French, derived lexemes are formed using three different processes, namely: suffixation, prefixation, and conversion.

Prefixation: concatenation of a segment, called a *prefix*, to the left boundary of the base’s stem (e.g., *POSSIBLE* → *IMPOSSIBLE*).

Suffixation: concatenation of a segment, called a *suffix*, to the left boundary of the base’s stem (e.g., *GÉANT* → *GÉANTISSIME*).

Conversion: creation of a new lexeme through a change in grammatical category, without any phonological change (e.g., *ORANGE_N* → *ORANGE_{Adj}*).

2.1.3 Families and paradigms

The concept of a paradigm was first studied in the field of inflectional morphology.

Inflectional paradigm: structured set of wordforms, typically represented as a table, in which each cell value is a form that embodies a combination of grammatical features.

In French, nouns inflect according to a single grammatical feature: number. As a result, the French noun paradigm contains only two cells (Table 2.1). In contrast, the adjective paradigm, which inflects for gender and number, contains four cells (Table 2.2), and the verb paradigm, whose form varies by mood, tense, person, and number, contains fifty-one cells.

lexeme	SG	PL
GARAGE	<i>garage</i>	<i>garages</i>

TABLE 2.1: Word-form paradigm of the noun GARAGE

BEAU	SG	PL
M	<i>beau</i>	<i>beaux</i>
F	<i>belle</i>	<i>belles</i>

TABLE 2.2: Word-form paradigm of the adjective BEAU

Then morphologists began to seek to extend this idea of a paradigm to derivational morphology. Extending this concept to derivation raises a difficulty specific to this subfield: derivational relationships are not obligatory, we do not have a fixed set of features to define the cells of a paradigm, unlike in inflection. It is this difficulty that is addressed, each in its own way, by the models proposed by Bonami and Strnadová (2019) and by Hathout and Namer (2022) and Namer and Hathout (2026). According to Bonami and Strnadová (2019), a derivational paradigm is a set of morphological families that are fully aligned with one another, where each family is viewed as a complete graph and where the cells of the paradigm may contain not a single lexeme but a set of units.

Derivational family (or morphological family): structured set of lexemes linked in pairs by a derivational relationship (e.g., AGRÉABLE, DÉSAGRÉABLE, and AGRÉABILITÉ are part of the same morphological family).

In other words, we take several families that have the same internal structure (the same semantic roles) and map them to one another. An example of a derivational paradigm is shown in Figure 2.3.

family	verb	action noun	agent noun
1	LAVÉ	LAVAGE	LAVEUR
2	SALÉ	SALAGE	SALEUR
3	COIFFÉ	COIFFAGE	COIFFEUR

TABLE 2.3: An example of French derivational paradigm, adapted from Bonami and Strnadová (2019)

LAVÉ is aligned with SALÉ, since both are verbs, LAVAGE is aligned with SALAGE, since both are action nouns, and so on. The paradigm is the set of these aligned families: the entire table.

This model has the merit of transposing the framework of inflectional paradigms very directly to derivation, by relying on contrasts in content (semantic contrasts)

between families rather than on contrasts in form. Hathout and Namer (2022) start from this same intuition but introduce an additional, more refined distinction: instead of a single entity, they distinguish the morphological paradigm from the derivational paradigm.

Morphological paradigm: homogeneous superposition of families that share both the same contrasts in form and the same contrasts in meaning. Hathout and Namer (2022)

Derivational paradigm: superposition of several morphological paradigms that are heterogeneous in form but homogeneous in meaning (for example, grouping the *-age/-eur*, *-ation/-ateur*, and *-ment/-eur* families under a single semantic pattern: verb / action / agent). Hathout and Namer (2022)

This makes it possible to clearly distinguish between formal regularity and semantic regularity at two distinct levels and to account for both the semantic consistency of derivational paradigms and the variation among the competing suffixes that realize them.

2.1.4 Stem and stem-space

When we discussed prefixation and suffixation, we used the term “stem” without really defining it. In this thesis, we distinguish between two types of stems: derivational stem and morphophonological stem.

Derivational stem: graphical sequence of the derivative corresponding to the abstract label denoted by X, which identifies a derivational alternation pattern between two lexemes (e.g., LENTILLE / LENTICULAIRE $\Rightarrow$ X / Xaire, where *lenticul* is the derivational stem).

Morphophonological stem: sequence of phonemes shared by all the word-forms within the same paradigm (e.g., the morphophonological stem of LENTILLE and that of LENTICULAIRE are, respectively, /lãtij/ et /lãtikyl/).

If we return to the concept of the inflectional paradigm discussed earlier, we can see that the stem (in the phonological sense) associated with each cell of the adjective ROUGE (Table 2.4) is /ruʒ/.

ROUGE	SG	PL
M	<i>rouge</i>	<i>rouges</i>
F	<i>rouge</i>	<i>rouges</i>

TABLE 2.4: Wordform paradigm of the adjective ROUGE

But for BEAU, the stem is either /bo/ (*beau, beaux*) or /bɛl/ (*belle, belles*). The former is found in the derivative BEAUTÉ, while the latter is found in the derivative EMBELLIR. It would seem that, depending on the adjective, different possible stems are used to create new lexemes. This type of phenomenon led Bonami and Boyé (2003) to propose the concept of stem-space (or thematic space).

Stem-space: indexed collection of stems.

Unlike the paradigm, which applies to a lexeme and describes the forms in which that lexeme occurs, the stem-space applies to an entire grammatical category: it does not describe the stems actually used by a given lexeme, but rather the number of slots needed to account for the full range of variation observed across the entire category. Thus, Bonami and Boyé (2003) first determined the size of the French verbal stem-space and concluded that it comprised 12 slots. This does not mean that each verb employs 12 different stems. In fact, the authors conducted a systematic study of all French verbs and concluded that 12 possible stem slots were needed to cover all cases of French verb conjugation. For more regular verbs (e.g., among others, verbs in the first group such as *NAGER*), the stems are almost all homomorphic (Table 2.5).

slot 1 ind.imp + ind.pst 1&2pl	slot 2 ind.pst 3pl	slot 3 ind.pst sg	slot 4 part.pst	slot 5 imp.pst sg	slot 6 imp.pst pl
naʒ	naʒ	naʒ	naʒ	naʒ	naʒ
slot 7 subj.pst sg + 3pl	slot 8 subj.pst 1&2pl	slot 9 inf.pst	slot 10 ind.fut + cond.pst	slot 11 ind.past + subj.past	slot 12 part.past part.past
naʒ	naʒ	naʒe	naʒə	naʒa	naʒe

TABLE 2.5: Stem-space of the verb *NAGER*

On the other hand, a verb like *BOIRE* fills these 12 cells with stems that vary greatly depending on the cell (Table 2.6).

slot 1 ind.imp + ind.pst 1&2pl	slot 2 ind.pst 3pl	slot 3 ind.pst sg	slot 4 part.pst	slot 5 imp.pst sg	slot 6 imp.pst pl
byv	bwav	bwa	byv	bwa	byv
slot 7 subj.pst sg + 3pl	slot 8 subj.pst 1&2pl	slot 9 inf.pst	slot 10 ind.fut + cond.pst	slot 11 ind.past + subj.past	slot 12 part.past part.past
bwav	byv	bwa	bwa	by	by

TABLE 2.6: Stem-space of the verb *BOIRE*

Bonami and Boyé (2003) subsequently extended this concept to other categories, such as adjectives. The stem-space for adjectives in French consists of two slots, whether we consider a regular adjective like *ROUGE* (which fills both slots with the same stem /ruʒ/) or an irregular adjective like *BEAU* (which fills these same two slots with two different stems, /bo/ and /bɛl/). This concept, which was initially theorized in the context of inflection, has been extended to word formation. In fact, Bonami, Boyé, and Kerleroux (2009) observed that certain slots in the stem-space served both inflection and word formation. Following this observation, they focused on word observation and discovered a thirteenth stem for the thematic space of verbs that would be used exclusively in derivation. This stem comes from the Latin supine and is used to form derived words formed by specific word formation processes, e.g., /resept/ is found in *RÉCEPTACLE* or *RÉCEPTIF*, both derived from *RECEVOIR*. Tribout (2010) subsequently proposed adding an additional stem (stem zero), used specifically in phenomena of conversion. However, these stems 0 and 13 are debatable.

2.1.5 Allomorphy and suppletion

Consider the following word pairs: GLACE/GLAÇON, FLEUR/FLORAL, POUDRE/PULVÉRISER, CHEVAL/HIPPIQUE, EAU/ÉVIER. From one pair to the next, the formal distance between the stems of the related words varies increasingly:

- we find the stem /glas/ in GLAÇON,
- the stem /flor/, although different, is relatively close to the stem /flœr/ (they share most of their phonemes),
- this is less and less the case between /puɔr/ and /pylver/,
- and this is no longer the case at all between /ʃəval/ and /ip/, or between /o/ and /ev/, which do not share any phonemes.

In the literature, these phenomena fall under the concepts of allomorphy and suppletion.

Allomorphy: situation in which several complex lexical forms are derived from different stems of the same base (e.g., PIANOTER and PIANISTE are formed, one from the stem /pjanɔt/, and the other from the stem /pjan/ of the noun PIANO). Bonami, Boyé, and Kerleroux (2009)

Suppletion: situation in which there is no formal link between the stems of two morphologically related lexemes (e.g., ÉQUESTRE and HIPPIQUE are related to CHEVAL, whose stem /ʃəval/ is not recognizable in either of the two adjectives). Bloomfield (1926), Boyé (2006), Carstairs (1990), Corbett (2007), Corbett (2009), Dressler (2005), Dressler (2015), Kiparsky (1996), Maiden (2004), Mel'čuk (1994), Moskal (2015), Tranel (1981), and Veselinova (2013)

The relationships between PIANISTE, PIANOTER and PIANO remain relatively easy for speakers to identify. In contrast, associating ÉQUESTRE or HIPPIQUE with CHEVAL is much less obvious and generally requires prior knowledge of this morphological relationship. In fact, these two concepts have in common the fact that the stems present in the two morphologically related lexemes are different. But where should the line be drawn between allomorphy and suppletion?

In fact, the problem lies in the fact that **formal distance** is not a binary distinction, it is therefore not possible to draw a clear line between allomorphy and suppletion based on this criterion alone. The criterion of formal distance involves viewing things gradually. Dressler (1985) therefore proposes two suppletions:

- strong suppletion (high formal distance),
- and weak suppletion (formal distance is low but not zero).

In addition to this notion of formal distance, Mel'čuk (1994) proposes introducing a new criterion: **reproducibility**. If variation is regular (that is, if it occurs in many pairs of morphologically related lexemes), then it is allomorphic; otherwise, it is suppletive. Thus, both authors would agree to analyze the pair FLEUR/FLORAL as a case falling between weak suppletion and allomorphy, but for different reasons: one emphasizes the regular nature of the alternation /œ/ ~ /o/ in French (since it is found in numerous pairs such as CHŒUR/CHORAL, DANGEREUX/DANGÉROSITÉ,

POREUX/POROSITÉ, *etc.*), while the other highlights the small formal distance between /floer/ and /flor/. Furthermore, according to Montermini (2010) and Montermini (2018), formal distance alone would not be sufficient to account for the degree of ease with which speakers establish a connection between morphologically related lexemes despite stem variation. This is why Lignon et al. (2025) propose addressing stem variation by combining several criteria:

- Formal distance
- **Integrity:** not all changes to a stem affect the perception of the morphological link in the same way; changes at the boundary between the stem and the affix are easier to identify than those affecting the core of the stem (e.g., in /abriko/ → /abrikotje/, the addition of /t/ occurs at the boundary, largely preserving the stem; DIABLE/DIABOLIQUE, the insertion of /o/ within the stem makes the connection less transparent)
- **Series effect:** the relative frequency of a variation in the lexicon (a regularly observed alternation is more easily recognized)
- **Family effect:** the distribution of a stem in its derivational family (the more often it appears in related lexemes, the easier it is to establish connections)

Other criteria are also mentioned in the literature, such as the etymological criterion for identifying different types of suppletion, which is not sufficient for Mel'čuk (1976), not necessary for Rudes (1980), and peripheral for Dressler (2015) or Hinzelin (2011). Other criteria could also be considered. One might think of word frequency in a corpus, which has the advantage of being easily measurable. The influence of spelling could also play a role: for example, the addition of the phoneme /t/ in /abrikotje/ is undoubtedly less surprising to the speaker since this consonant is already present in the spelling of ABRICOT. Finally, the familiarity of lexemes could also come into play. The more familiar and frequently encountered a word is, the more likely it is to be easily linked to its base. Thus, the connection between LECTEUR and LIRE is likely easier to establish than that between ONOMASTIQUE and NOM, since ONOMASTIQUE is a less common word.

In addition to all this, many authors view suppletion as exclusively associated with the inflectional domain (Hacken and Panocová, 2020). Consequently, relationships in which the members are formally distant (such as CHEVAL/HIPPIQUE) are not derivational but rather lexical.

To summarize, although suppletion is widely theorized in inflection (*cf.* Bonami and Boyé (2003), who, through their concept of stem-space, highlighted the suppletive stems of the verb ALLER: /ir/ in *ira, iront*, or /al/ in *allons, allèrent*), all linguists discuss this notion of suppletion, everyone uses it, but there is no consensus regarding this notion in derivation.

2.2 Lexical resources

We will now present in detail the main resources used in the broader project of which this work is a part, as outlined in the introduction (Section 3.2.1). We focus on the following derivational and lexical resources: Démonette, GLeMM-3, Phononette-1 & 2, Kaikki, and the etymological dictionary known as FEW (Französisches Etymologisches Wörterbuch).

rid	lemme_1	lemme_2	type_cstr	patron_deriv	complexite	orientation
r13962	boursouffler _v	boursoufflure _{Nr}	suf	X/Xure	simple	as2des
r13959	boursouffler _v	boursoufflage _{Nm}	suf	X/Xage	simple	as2des
r63185	repasser _v	repasseuse _{Nr}	suf	X/Xeuse	simple	as2des
r63168	repassage _{Nm}	repasseuse _{Nr}	suf	Xage/Xeuse	simple	indirect

FIGURE 2.2: Table of relations (sample)

The second table (Figure 2.2) provides various information regarding the relationship between pairs of lexemes in the database, including:

- Relation identifier (e.g., r13962)
- Type of construction (e.g., BOURSOUFFLURE is a suffixation)
- Derivational pattern (e.g., BOURSOUFFLER/BOURSOUFFLURE $\rightarrow$ X/Xure)
- Complexity (e.g., simple)
- Orientation (e.g., BOURSOUFFLER $>$ BOURSOUFFLURE has the orientation ancestor-to-descendant)

We can note that we immediately encounter the concept of a derivational family introduced in Section 2.1.3. Furthermore, derivational relations (Figure 2.2) not only link base words (REPASSER) to a direct derivative (REPASSEUSE or REPASSAGE) but also derivatives to one another (REPASSAGE, REPASSEUR) (when the relation is labeled “indirect”), which allows us to represent the family as a graph rather than as a simple list of base-derivative pairs. In Chapter 3, we take a closer look at the aspects of *complexity* and *orientation*.

2.2.2 GLeMM-3

GLeMM-3 is a very large morphophonological resource that provides formal and semantic descriptions of derivational relationships (Hathout et al., 2026). Its large size is partly due to the fact that it covers seven languages: German, English, Spanish, French, Italian, Polish, and Russian, with an identical format and annotation for each of them, which allows the same processing to be applied to all covered languages. More specifically, GLeMM-3 is a multilingual collection of comparable derivational lexicons. This resource is built from the various language-specific editions of the Wiktionary³ (e.g. the Wiktionnaire⁴ for the French GLeMM-3), using structured electronic dictionaries (GLAWI; Sajous and Hathout, 2015; Hathout and Sajous, 2016, ENGLAWI; Sajous, Calderone, and Hathout (2020), GLAW-IT; Calderone, Sajous, and Hathout (2016)) or *via* data extracts from the Kaikki project (Ylonen, 2022), a project presented in Section 2.2.4.

GLeMM-3 consists of two tables per language: *pairs.tsv* and *defs.tsv*. Pairs of candidate lexemes are automatically identified and annotated by FAPinette⁵, a system based on the detection of formal analogies between pairs of lemmas (the Lepage

³See <https://www.wiktionary.org/>

⁴See <http://fr.wiktionary.org>

⁵See <https://src.koda.cnrs.fr/nabil.hathout.1/fapinette/-/tree/master>

(1998) method adapted by Hathout et al. (2020)): if a pair of lemmas forms formal analogies with a sufficient number of other pairs, its morphological relationship is considered regular, and the pair will be included in the *pairs.tsv* table. In total, the corpus contains more than 2.3 million lexeme pairs across all languages combined.

language	lemma 1	lemma 2	stem	exponent 1	exponent 2
Spanish	rastrero _{Adj} 'despicable'	rastreramente _{Adv} 'despicably'	rastrer	^(.+)o\$	^(.+)amente\$
English	sensitivity _N	electrosensitivity _N	sensitivity	^(.+)s\$	^electro(.+)s\$

TABLE 2.7: Examples of entries of the *pairs.tsv* table, adapted from Hathout et al. (2026)

Table 2.7 provides an example of what can be found in this table in English and Spanish, among other languages:

- Lemma 1, which forms the base of the analogically formal relation (e.g., sensitivity)
- Lemma 2, which forms the derivative of the analogically formal relation (e.g., electrosensitivity)
- Stem (e.g., sensitivity)
- Exponent patterns (exponent 1 and exponent 2) associated with each lemma (e.g., for “electrosensitivity”, its exponent pattern is ^electro(.+)s\$)

language	lemma 2	definition 2
Spanish	rastreramente _{Adv}	de manera rastro 'in a vile way'
English	electrosensitivity _N	the sensitivity to electricity

TABLE 2.8: Examples of entries of the *defs.tsv* table, adapted from Hathout et al. (2026)

The second table, *defs.tsv*, is a subset of the previous one that adds a morphological definition of the second lexeme (definition 2) based on the meaning of the first (for example, ELECTROSENSITIVITY = ‘the sensitivity to electricity’), which allows us to indicate the position of the second lexeme relative to the first within their paradigmatic family. Table 2.8 shows an excerpt from this table for Spanish and English. In total, more than 700,000 pairs are provided with a morphological definition.

2.2.3 Phononette-1 and Phononette-2

The third resource, Phononette (Namer, Hathout, and Lignon, 2017; Lignon et al., 2025), is a direct continuation of Démonette. This resource is specifically designed to study stem variations and contribute to the allomorphy *versus* suppletion debate (Section 2.1.5). It merges Démonette’s LT and RT and calculates, for each derivational relation, the formal distance between the phonemic transcriptions of the stems of the two related lexemes.

Of all the relations in Démonette, only those that actually exhibit variation were included in the Phononette-1 test version (Namer, Hathout, and Lignon, 2017), amounting to 19,032 derivational relations, of which 34.3% (6,540 entries) were manually annotated.

couple	transcription	patron	type of variation	frequency
<i>stable/stabil-ité</i>	[stabl] / [stabil]	Xbl/Xbil	adjunction of a vowel	33.16%
<i>bonbon/bonbonn-ière</i>	[bɔ̃bɔ̃] / [bɔ̃bɔ̃n]	Xɔ̃/Xɔ̃n	denasalization	29.48%
<i>cheval/hipp-ique</i>	[ʃəval] / [ip]	X/Y	complete stem suppletion	11.41%

TABLE 2.9: Classification and frequency of stem alternation types in the manually validated Phononette-1 sample, adapted from Namer, Hathout, and Lignon (2024)

The authors propose a classification of these variations, an excerpt of which is presented in Table 2.9. For example, let’s take the pair STABLE/STABILITÉ (row 1). The morphophonological stems of the base and the derivative are, respectively: [stabl] and [stabil]. We observe that these two stems are not identical, and this variation is highlighted in the (morphophonological alternation) pattern Xbl/Xbil. This variation simply consists in the addition of a vowel, which accounts for 33.16% of the variations in the corpus. The main advantage of Phononette is that it offers, beyond mere formal distance, additional criteria derived directly from the paradigmatic framework: stem integrity, the series effect, and the family effect, criteria that were presented in Section 2.1.5. This work was carried out in the second version of Phononette (Phononette-2) based on 90,845 entries from Démonette that were morphophonologically annotated.

2.2.4 Wikextract and the Kaikki project

Developed by Ylonen (2022), Wiktext⁶ is a tool that converts the content of Wiktionary articles (across all language editions) into machine-readable data in JSON format. Unlike previous extractors, Wiktext fully interprets and expands the templates and modules used by Wiktionary contributors, enabling it to perform a more comprehensive and robust extraction of headwords, inflected forms, translations, pronunciations (with links to audio files), and above all, as far as we are concerned, etymological sections.

Each extracted entry is a JSON object associated with a word, a language (lang), a part of speech, as well as an “etymology_text” section and a list of meanings (senses). The data is regularly updated on the website <https://kaikki.org/>.

2.2.5 FEW

The FEW is a reference dictionary of etymology and history covering all the Gallo-Romance languages (French, the Oïl dialects, Franco-Provençal, Occitan, and Gascon). Its compilation began in 1922 under the direction of the Swiss philologist Walther von Wartburg. It was written primarily in German until 1980, and thereafter in French. For each etymon, the work traces its origin, history, and all derivatives attested in the written and spoken Gallo-Roman varieties between the 9th and 20th centuries. It comprises a total of 31 volumes and nearly 17,000 pages, making it one of the most extensive lexicographical works. Since 1993, following an agreement between the Swiss National Science Foundation and the CNRS, the FEW’s editorial center has been housed at the ATILF, where its digitized version is now freely available online⁷.

⁶See <https://github.com/tatuylonen/wiktext>

⁷See <https://lecteur-few.atilf.fr/>

Unlike the resources presented earlier (Démonette, GLeMM-3, Phononette), which describe synchronic derivational relationships between lexemes, the FEW adopts a diachronic perspective: each entry groups together all the units that are formally and semantically derived from the given etymon, regardless of how far apart they are synchronically from their base. The organization by etymon makes the FEW relevant to the work carried out in this thesis: the suppletive stems that BREF aims to record are, by definition, those whose connection to their base has been severed from the perspective of synchronic derivation, but which remain linked to it through a common etymon. In this thesis, the FEW serves as a primary source for verifying and documenting this type of etymological relationship. In the longer term, we hope to conduct a pilot study using this resource to fully exploit the data and ensure that it complements the data to be extracted from Kaikki.

Chapter 3

Build BREF

This chapter describes the experimental approach adopted to develop the intermediate dataset used to create BREF. This intermediate dataset contains a wealth of information provided by the LLM, and the goal is to create a dataset that includes all of this information for as much data as possible. We detail the choices made regarding the construction of the context, the prompting strategy, and the development of a reference dataset that enables the evaluation of the models' performance.

3.1 General methodology

As introduced in Chapter 1, the general idea behind this internship project is to build a lexicon that associates each lexeme with its suppletive stems, as distinguished from its inflectional stems. For example, for CHEVAL, the lexeme contains two inflectional stems: /ʃəval/ and /ʃəvo/. In contrast, the forms /ek/ (e.g., ÉQUESTRE, ÉQUITATION, ÉQUIDÉ), /ip/ (e.g., HIPPIQUE, HIPPISE, HIPPOPOTAME), or /kaval/ (e.g., CAVALIER, CAVALERIE, CAVALER) are not part of this inflectional system: they are suppletive stems used only in certain derivatives. In fact, we consider any stem to be a suppletive stem if its form is directly inherited or borrowed from an ancient etymon (primarily Latin or Ancient Greek) and has remained close to that etymon (ÉQUESTRE, comes from the Latin *equus* 'horse', in which we find the same /ek/ sequence), whereas inflectional stems diverge from it (CHEVAL comes from the Latin *caballum*, in which we do not find the same sequence /ʃəval/ versus /kabal/). To recover the so-called suppletive stems, we will: 1) draw on the etymological information from Kaikki, 2) use LLMs, and 3) perform human validation.

3.1.1 Why use Wiktionnaire data by means of the Kaikki database?

As presented in Section 2.2.4, Kaikki gives structured, computer-processable access to the etymological sections of Wiktionary entries, precisely the kind of information (etymon, source language, affix) needed to build our database. Example 1 illustrates, using the lexeme DANSE, the type of etymological description that we can find in Wiktionnaire.

- (1) DANSE → Déverbal de *danser*, issu d'une racine romane qui signifie « tirer en longueur » apparenté au latin *tendere* (« tendre »), *tensus* (« tendu ») ; danse étymologiquement désigne une chaîne, une file. L'allemand présente *Reigen* (« danse » , *Tanz* est un emprunt postérieur aux langues latines) et *Reihe* (« file, rangée »).

The version of Kaikki used in this study is the one dated 03-03-2026¹, which was the most recent at the time the project was launched. This database was chosen

¹See the last version here: <https://kaikki.org/dictionary/rawdata.html>

over other resources, notably the TLFi² (Trésor de la Langue Française informatisé) and the FEW, precisely because of its structure (see Chapter 6, which discusses the sources that are more difficult to use). Wiktionary remains a subject of controversy: as we know, anyone can contribute to it and, consequently, write the etymological section for a lexeme. According to Sagot (2017), the etymological information in Wiktionnaire can nevertheless be considered generally reliable: it often reflects the most recent scientific studies that represent a consensus, which is frequently cited in the references.

3.1.2 Why using LLMs?

To extract data from Kaikki, we have used LLMs. The principle is simple: we provide the model with a context that we know contains the information we are looking for, and then we ask it questions, in the form of a prompt, to guide its morphological analysis. This context will be obtained by selecting different parts of the Kaikki input corresponding to the lexeme under study (see Section 3.2). It is important to emphasize that we intend to use an LLM as an extraction tool and avoid, as much as possible, having it reason based on its own knowledge. In fact, by constraining the model to a clearly defined context, we reduce the probability that it will draw on external knowledge, thereby limiting errors and hallucinations. This approach allows us to maintain significant control over the extracted information. The alternative approach for extracting etymological stems from the Kaikki data would have been a rule-based method. However, this method has its limitations. Indeed, etymological data is written in natural language and exhibits significant variability in the presentation of information. As an illustration, let's consider the etymological descriptions of the three lexemes in examples 2, 3 and 4.

- (2) POLYADELPHE → Mot dérivé de adelphe (« frère »), avec le préfixe poly- (« plusieurs »).
- (3) PANCALISME → Du grec ancien *πάκαλος*, *pankalos* (« tout beau ») et -isme.
- (4) ÉVIER → euwier. Eve est une variante du XIème siècle de eau, à rapprocher avec son équivalent poitevin-saintongeais ève/aeve ou wallon aiwî. Évier désigne le récipient qui contient l'eau, comme aiguère dont il est le doublet naturel (voir aigue). Il provient sans doute du latin *aquarium* ou même du latin *aquarius* qui aurait pu devenir un adjectif en lieu et place de latin *aquatique*.

As we can see, the first two descriptions more or less follow the pattern “[...] *etymon* (« meaning of the etymon »), *affix*”, but the last one does not follow it at all. Extracting stems using rule-based methods such as regular expressions would not allow us to handle this variability. Rule-based methods have the advantage of being interpretable. In other words, when a result is obtained, it is possible to trace the reasoning that led to that output and understand its various steps. They thus offer total control over the decision-making process and the results produced. In contrast, LLMs are black-boxes. It is not possible to precisely reconstruct the internal reasoning that leads to a given response, which limits their interpretability. In this context, human validation is essential to assess the quality of the results produced as well as their linguistic relevance. This oversight is all the more critical given that LLMs can generate biases or unpredictable responses, making a critical review of their outputs necessary (Ouyang et al., 2022; Ji et al., 2023).

²See <http://www.atilf.fr/tlfi>

3.2 Pipeline for the intermediate dataset

Before designing the final pipeline to create the intermediate dataset, we conducted a series of experiments on the Kaikki data. These tests had three objectives: 1) to determine the information to provide to the LLM, which we will refer to as context, 2) to identify the most appropriate questions to ask it in order to obtain a relevant morphological analysis, and 3) to get an idea of the prompt and model to use.

3.2.1 Establishment of the context

lemma	cat	def	etym	fam
oblation	N	Offrande, action par laquelle on offre quelque chose à Dieu.	Du latin oblatio (« offrande »).	oblationnaire, oblat, offreuse, oblatif

TABLE 3.1: An example of context

The context was built up gradually, as the experiments progressed. Each new piece of information was added when it helped improve the quality or reliability of the analyses produced. The final context is illustrated by the entry OBLATION in Table 3.1. It contains five pieces of information, deemed relevant for a morphological analysis, which we will explain below.

The most basic piece of information is, of course, the lexeme itself (Table 3.1, column “lemma”), or at least one of its graphic forms, which is the subject of morphological analysis. However, this information alone has proven to be insufficient. We have therefore added the lexeme’s part of speech (Table 3.1, column “cat”). Indeed, French contains many homographs, and the part of speech helps resolve some of the ambiguities by correctly guiding the analysis, particularly from an etymological perspective. For example, the lexemes LIRE_N and LIRE_V , although they share the same graphic form, do not have the same etymological origin. The noun comes from the Italian *lira*, itself derived from the Latin *libra* ‘pound, the unit of weight’, while the verb is inherited from the Latin *lĕgĕre* ‘to read, literally to gather’. The expected etymological stems are therefore different: *leg-* for the verb, and *lir-* or *libr-* for the noun. However, grammatical category alone does not always allow us to distinguish between homonymous lexemes belonging to the same grammatical class. We have therefore incorporated a third piece of information: the definition of the lexeme (Table 3.1, column “def”). This allows us to precisely identify the meaning in question and, consequently, to select the correct etymological description. For example, we do not analyze the etymological stems of AVOCAT_N ‘profession’ “From the Latin *advocatus* (‘defender, advocate’) in the same way.” and AVOCAT_N ‘fruit’ “From the English *avocado*, which itself comes from the Spanish *aguacate* (1640), derived from Classical Nahuatl *ahuacatl* (with the same meaning).” The definition allows us to precisely identify the lexeme in question and access the correct etymological description. Once the lexeme has been identified, the most important information for our project is indeed the etymological description of the lexeme (Table 3.1, column “etym”). The latter generally provides the necessary information regarding the suppletive **stem** (for derivatives that contain one), the derivational **process**, or even the **affix** itself, as is the case with the etymological description of LACTIQUE (5).

(5) $\text{LACTIQUE} \rightarrow$ Du latin **lac**, **lactis** (« lait ») avec le **suffixe** **-ique**.

Of all the information extracted from Kaikki, the etymological description therefore constitutes the central element of the context provided to the LLM. Other information, such as the part of speech and the definition, primarily serve as checks and validations for the human user. During the experiments, we also chose to add to the context lexemes belonging to the same derivational family as the lexeme under study (Table 3.1, column “fam”). The goal is to enable the LLM to identify other lexemes built on the same stem and, ultimately, to study the productivity of suppletive stems within derivational families. To build these families, we used several categories from the Kaikki dataset, notably *derived*, *paronyms*, *synonyms*, *related*, *antonyms*, and *holonyms*. These categories contain many relevant lexemes, but also a significant amount of noise. For example, for the lexeme SIÈGE, we find words belonging to the same family, such as “assiéger” and “siéger”, but also numerous expressions like “arme de siège”, “bain de siège”, “batterie de siège”, or “engin de siège”, which are not derived lexemes. Data preprocessing was therefore necessary to limit this noise. We have removed expressions consisting of multiple words, as well as words containing a hyphen, such as “bras-le-corps” or “mont-de-piété”. Finally, we have supplemented this information with the derivational families from Démonette. This resource provides an initial set of lexemes belonging to the same family. The data extracted from Kaikki serves as a supplement, by providing additional lexemes sharing the same stem that may be absent from Démonette, and can thus complement the latter. For example, the derivational family of the lexeme OBLATION in Démonette contains the following lexemes: OFFREUSE, OFFREUR, OFFRANT, OFFRIR, OFFRANDE, OBLATEUR, OBLATIVITÉ, OBLATIF, whereas Kaikki provides another one, OBLATIONNAIRE.

3.2.2 Questions asked to the model

Given this context, we want the LLM to extract information about lexemes that have undergone derivation. More specifically, we want it to extract the most important piece of information for constructing BREF: the stem. The first question we ask, therefore, is what the derivational process is. This leads to several possible scenarios.

- The first: there is no process, and thus no derivation; in this case, we do not need to analyze this word, and the following questions will not apply to it.
- The second: there is a process, but it is not affixation; in this case, it is of no interest to us (Section 3.3 details what is and is not of interest to us within the scope of this study).
- The third case: we are dealing with a case of affixation, in which case, to find the stem, we must know the affix.

Indeed, information about the stem is never explicitly mentioned in the etymological description, unlike the affix (see the etymological description of LACTIQUE under (5)); therefore, the LLM will simply have to deduce it by performing the following operation: stem = entry - affix. Thus, logically, the second question consists of asking for the affix, while being aware that this question would not make sense for the first and second scenarios. We now have all the information at our disposal to identify the stem: the third question we ask the LLM is therefore the value of this stem. Next, we want additional information about this stem. In fact, we are working on specific stems (the suppletive ones). To compile our lexicon of BREF stems, we wish to gather more information about each stem in order to better understand how

it works. Since we are examining stems from an etymological perspective, we have decided to trace the etymological history of the word back to a certain point. Indeed, Romance linguists distinguish between direct etymology (*etimologia prossima*) and distant etymology (*etimologia remota*) (Buchi, Baudinot, and Petrequin, 2005). Direct etymology constitutes the most recent stage (e.g., Latin *mātrēm* ‘mother’ constitutes the recent etymology of the French lexeme *MÈRE* ‘woman who has given birth to a child’), whereas the distant etymology corresponds to the earlier stages that are the subject of Indo-European etymology (e.g., the Indo-European *māt’er-* ‘mother’ constitutes the distant etymology of the French lexeme *MÈRE* ‘woman who has given birth to a child’). We have decided to focus on recent etymology, as it is more relevant to the morphological analysis of modern French, since the formal link between the derivative and its etymon is more transparent than with distant etymology. Since we are working with derivatives, it is highly likely that our recent etymology includes several etymons. Indeed, we can expect that our French derivative comes from a Latin or Ancient Greek unit that was itself already formed from a Latin or Ancient Greek stem. This is the case for the lexeme *STRABISME*, which, according to its etymological description (6), comes from the etymon *strabismos* ‘strabisme’, which is itself formed in Ancient Greek from the stem *strabos* ‘qui louche’.

- (6) *STRABISME* → Du grec ancien $\sigma\tau\rho\alpha\beta\iota\sigma\mu\acute{o}\varsigma$, *strabismos*, issu de $\sigma\tau\rho\alpha\beta\acute{\iota}\zeta\omega$, *strabizō* (« loucher »), de $\sigma\tau\rho\alpha\beta\acute{o}\varsigma$, *strabos* (« qui louche »).

So we are going to ask the LLM to extract all the etymons and their meanings, starting from the etymon which we will call the “direct” one (the word that is historically closest to the modern word) all the way back to its stem. This allows us to understand where the stem comes from, which etymon(s) it is modeled after, and its meaning in modern French. Next, based on this latest information, we try to have the system extract two additional pieces of information: the stem’s source language and its semantic base. What we call the semantic base is, in fact, the meaning of the stem in French. For example, the stem *card* in *CARDIAQUE* means ‘cœur’. It generally corresponds to the meaning of the last etymon that the LLM can extract (with *STRABISME*, this would be ‘qui louche’), but this is not always the case. Let’s take the lexeme *HUMAIN*. According to the etymological description given in (7), we can see that the last etymon the LLM can extract is *humus*, which means ‘terre’; however, *HUMAIN*, in the sense in which it was borrowed into Latin, comes from the meaning ‘homme’ and not ‘terre’ so the semantic base we expect is ‘homme’.

- (7) *HUMAIN* → Du latin *humanus*, dérivé de *homo* (« homme »), avec le suffixe *-anus*. Dérivé de *humus* (« terre »), avec le suffixe *-o*, *-onis*, *-inis*, littéralement « terrien » .

In other words, the information that the LLM is able to extract when we ask it to find all the etymons and their meanings allows us to understand the evolution of the stem, but does not necessarily allow us to deduce the semantic base: it is not necessarily the last etymon that interests us. Table 3.2 clearly summarizes the problem we encountered with *STRABISME* and *HUMAIN*: the base ‘qui louche’ does indeed come from the last etymon collected, while the base ‘homme’ comes from the second.

We must therefore rely on other means to access the semantic base of the stem, which is why we are addressing this question in isolation and separately. As for the source language, we expect the LLM to provide the language of the direct etymon. It is important to clarify this, since it is possible that when tracing the recent etymology of a lexeme, we may move from one language to another (borrowings, derivations using a foreign stem). For example, *CARNASSIER* originates from the

lexeme	etymology	etymons	base
STRABISME	Du grec ancien στραβισμός, strabismos, issu de στραβίζω, strabizô (« loucher »), de στραβός, strabos (« qui louche »).	strabismos, strabizô, strabos	‘qui louche’
HUMAIN	Du latin humanus, dérivé de homo (« homme »), avec le suffixe -anus. Dérivé de humus (« terre »), avec le suffixe -o, -onis, -inis, littéralement « terrien » .	humanus, homo , humus	‘homme’

TABLE 3.2: Summary of what we expect for STRABISME and HUMAIN

Occitan carnassièr (or carnacier) ‘qui se nourrit de chair crue’, which itself comes from caro, carn ‘chair’. In this specific case, we want the LLM to respond that the source language of CARNASSIER is Occitan. Finally, we ask one last question to obtain all the lexemes in the derivational family of the entry that share the same stem as it. For example, for OBLATION in Table 3.1 (column “fam”), the LLM must extract OBLATIONNAIRE, OBLAT, OBLATIF, but not OFFREUSE. This information is of particular interest to us since it will appear in BREF.

In summary, we ask the LLM seven different questions, namely: the derivational process (if any), the affix (if any), the stem, the etymon(s) and their semantic equivalent(s), the source language, the semantic base, and the lexemes in the derivational family that share the same stem as the entry.

3.2.3 Prompt elaboration

To ask these questions, we use a prompt. Writing a good prompt is extremely important because it affects a model’s performance. It has been shown that to write a good prompt, one must be specific by using long, detailed instructions, requesting a specific output (such as JSON, which is an easy-to-parse format familiar to LLMs), or by providing examples (few-shot prompting) (Tony et al., 2021; Liu et al., 2023). According to Leidinger, Rooij, and Shutova (2023), the linguistic formulation of prompts also has a measurable impact. Using a particular grammatical mode (imperative, interrogative, declarative), a specific verb tense, a certain modal verb (“should”, “must”), or specific lexical choices (one synonym rather than another) does not have the same impact on a model’s performance. This study also points out that good prompts are not universal: a prompt that is effective for one model may be mediocre for another. However, detailed instructions should not result in unnecessarily long prompts. Indeed, Liu et al. (2024) highlight the lost in the middle phenomenon: in a long prompt, LLMs make better use of information placed at the beginning or the end of the prompt than that located in the middle. It is therefore necessary to strike a balance between the richness of the instructions and their conciseness.

In practice, to find the best prompt, we try one, analyze it, and repeat the process until performance stabilizes. To establish a starting point for our methodology, we conducted several small-scale tests in which we compared different models,

prompts³, and prompting methods. Among other things, we wanted to gain insight into the effectiveness of few-shot learning and the ideal level of detail for the prompt. As a reminder, few-shot is a prompting technique that involves providing the model with a few example responses to guide it in its task. This technique relies on the ability of LLMs to generalize from a limited number of examples provided in context. We therefore hypothesized that this approach could improve the results obtained for our task. We tested this technique, taking care not to provide more than five examples. Indeed, several studies show that adding additional examples does not necessarily lead to improved performance and may even, in some cases, reduce the model’s performance (Brown et al., 2020; Tang et al., 2025; Perez, Kiela, and Cho, 2021). We drew several conclusions from these small-scale tests: 1) few-shot training does not seem to yield a noticeable improvement, 2) a detailed prompt seems to encourage the model to invent information when the answer is not in the context, 3) it appears that requiring a JSON output format increases the number of parsable responses. However, our level of confidence in these observations remains relatively low, as the primary purpose of these tests was to help us become familiar with the methodology rather than to rigorously determine the optimal prompt. A more in-depth study of our benchmark dataset would be necessary to confirm these hypotheses. In particular, it would be worthwhile to explore the various biases associated with few-shot prompting described in the literature, such as the influence of the order of examples, majority label bias, or recency bias (according to which the most recent examples or instructions may have a disproportionate influence on the model’s response) (Tony et al., 2021).

3.3 Construction of a reference dataset

We have built a reference dataset to evaluate the responses of the LLM. Based on previous tests, we know what information is needed in the context and what questions we want answers to. The dataset size is set at 100 lexemes, divided equally between 50 non-derived lexemes and 50 derived lexemes corresponding to potential suppletions. Before explaining the method for extracting these 50 derived lexemes and 50 potential suppletions, we need to clarify what we mean by “derived lexeme” and “non-derived lexeme”. The following explanations are based on the organization of lexical units in the lexicon shown in Figure 3.1.

For the purposes of this study, a derived lexeme is a morphologically complex lexeme that is not a compound. Compound lexemes are therefore excluded from our dataset. Among the derivational processes, we have chosen to focus on affixation, leaving conversion aside. This choice is based primarily on theoretical grounds. Conversion (see its presentation Section 2.1.2), by definition, produces no formal variation between the base and the derivative. Consequently, it cannot, *a priori*, give rise to a suppletive stem: this would imply that the base itself is already a suppletive stem, which is not possible since such stems do not occur in inflection. Conversion is therefore not relevant to our study of suppletion. However, there are some borderline cases that qualify this exclusion, but that is beyond the scope of this thesis. These cases are discussed in Chapter 6.

A non-derived lexeme, on the other hand, is a morphologically simple lexeme (e.g., *CHAT*, *PETIT*), as opposed to complex, whose internal structure comprises several morphological elements and which results from a process of lexical construction.

³See all the prompts in the Appendix A.1

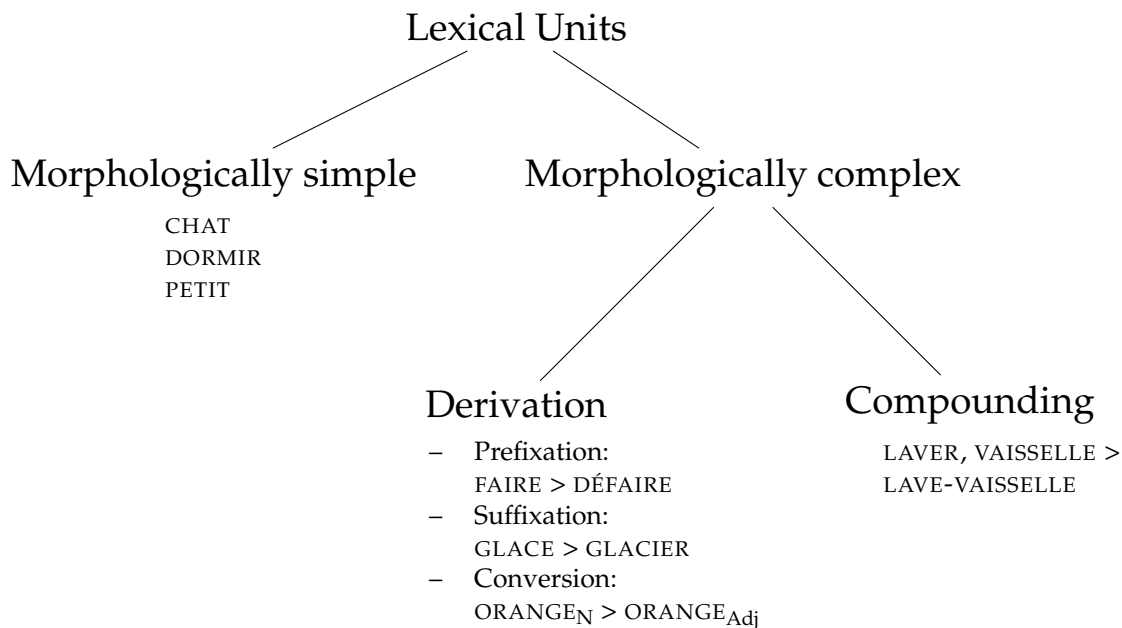


FIGURE 3.1: Morphological organization of the French lexicon

However, borderline cases complicate matters. Take the example of *FRIABLE*: in contemporary French, this adjective is simple, since there is no verb *FRIER* that could serve as its base. Its etymological description indicates, however, that it comes from the Latin *friare*, *frio* ‘s’effriter’, suffixed with *-bilis*. A lexeme is therefore considered complex depending on whether one adopts a synchronic or diachronic perspective. *FRIABLE* is a simple, non-derived lexeme in the synchronic sense, although its etymological description contains information about morphological processes and affixes, typical of a complex, derived lexeme (Aronoff, 1976). For the purposes of our study, we use the synchronic criterion to distinguish between our two classes of lexemes. A lexeme is therefore considered non-derived when it is not formed, in modern French, by prefixation or suffixation.

3.3.1 Selection of dataset entries

To select the suppletive forms (among the derived lexemes) in our dataset, we relied on *Démonette*. As a reminder (Section 2.2.1), this resource is structured into two main tables: the LT, which contains information specific to each lexeme, and the RT, which provides information about the complexity of a derivational relation between two lexemes, and its orientation. We selected lexeme pairs in which one is ascending (i.e., the derivational base) and the other is descending: the derivative. For example: *ABUS* > *ABUSIF*; *ABUSIF* (descending) is derived from *ABUS* (ascending). This is coded as *as2des* in *Démonette*. We then wish to identify a suppletive relationship, that is, a relationship in which the meaning of the derivative can be inferred from that of the base, but where the form of the derivative is not directly formed from its ascendant. Let’s take the pair *ÉCOLE* > *SCOLAIRE*. *SCOLAIRE* can be directly defined from the noun *ÉCOLE* and *SCOLAIRE* = ‘qui est relatif à l’école’, but the derivative is formed from the suppletive stem /skol/ of the base. This type of relationship is coded as *motiv-sem*. The combination of these two properties, which are characteristic of suppletive relations, allows us to obtain a list of candidate derivational relations

involving a derived word with a suppletive stem. Some examples of these pairs are presented in Table 3.3, taken from Démonette.

lemma 1	lemma 2	cat 1	cat 2	cstr 1	cstr 2	complexite <i>motiv-sem</i>	orientation <i>as2des</i>
offrir	oblation	V	N	X	Xion	motiv-sem	as2des
glande	glandulaire	N	Adj	X	Xaire	motiv-sem	as2des
grain	granulaire	N	Adj	X	Xaire	motiv-sem	as2des
lentille	lenticulaire	N	Adj	X	Xaire	motiv-sem	as2des
tube	tubulaire	N	Adj	X	Xaire	motiv-sem	as2des
valve	valvulaire	N	Adj	X	Xaire	motiv-sem	as2des

TABLE 3.3: Selection of suppletive lexemes from Démonette with *complexite:motiv-sem* and *orientation:as2des* parameters

Since the orientation is set to *as2des*, the ascending lexeme (the base) is in the column “lemma 1”, while the descending lexeme (the derivative) is in the column “lemma 2”. Since the complexity is set to *motiv-sem*, the descending lexeme (lemma 2) corresponds to a suppletive derivative. The columns “cat 1” and “cat 2” therefore refer, respectively, to the grammatical category of lemma 1 and to the grammatical category of lemma 2. All the relationships illustrated in this table are cases of suffixation. This is why, in the column “cstr 1”, lemma 1 is represented by an X, and column “cstr 2” contains the derivational pattern of lemma 2 relative to ‘X’. This table thus shows us that the suppletive pair OFFRIR_V > OBLATION_N (row 1) results from suffixation with *-ion*.

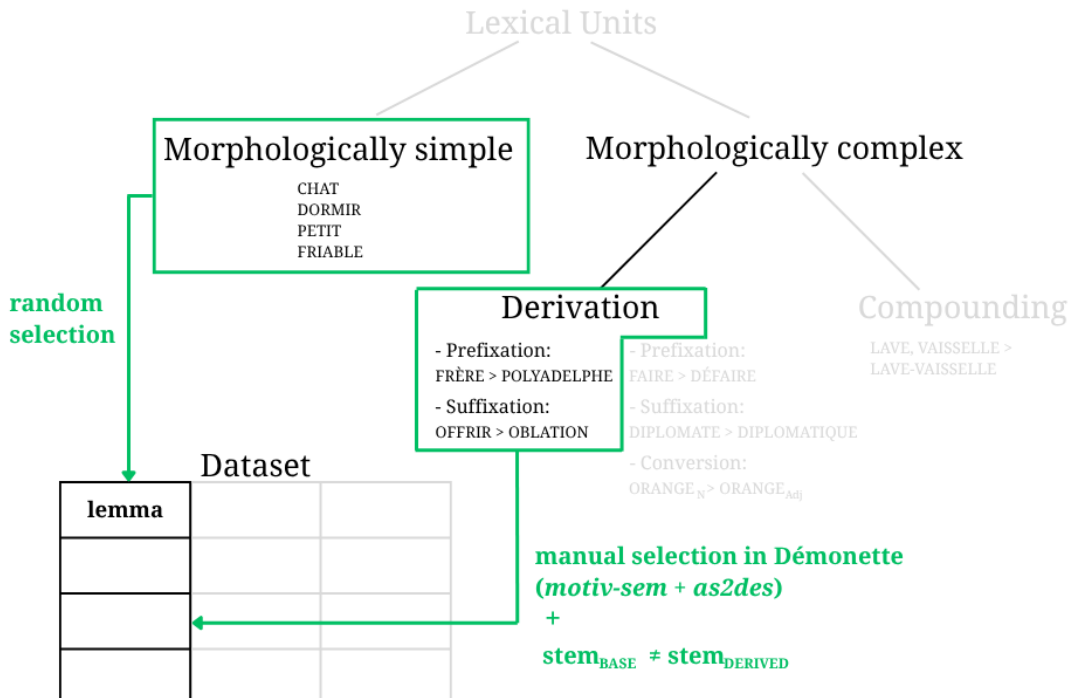


FIGURE 3.2: What is in my dataset?

The 50 suppletions were selected manually, while the 50 non-derived lexemes were chosen at random. Figure 3.2 summarizes the types of lexemes found in our dataset. For each lexeme, we identified its Kaikki entry based on its form and grammatical category. Non-derived lexemes (in the synchronic sense defined above) serve as

distractors: even though our goal is to study lexemes with a suppletive stem, that is, those resulting from derivation, we deliberately provide the LLM with a large amount of irrelevant data to study its behavior when processing such data. Indeed, in our dataset, we know which entries are derived and which are non-derived. However, when we apply our methodology on a larger scale to new entries, these may, in principle, be either derived or not. It is important to ensure that the LLM processes non-derived entries as we intend. Indeed, we initially expect that these distractors will not elicit any response from the model for questions relating to the formation process, the affix, the stem, etc. This expectation must, however, not be taken for granted: there is nothing to prevent a simple lexeme in synchrony from having, like *FRIABLE*, an etymological description that documents an ancient formation process. In such a case, an LLM's response is not necessarily an error, but may accurately reflect information that is actually present in the provided context. This nuance, which had not been anticipated when the dataset was created, is addressed in the qualitative analysis of the results (Section 4.1.2).

proc	aff	rad	etym_mean	lang	base	etym_fam
suffixation	-ion	oblat-	oblatio, 'offrande'	latin	offrir	oblationnaire, oblat, oblatif

TABLE 3.4: Questions and expected answers (sample)

The reference dataset consists of the context part shown in Table 3.1 and the question-answer part shown in Table 3.4. We annotated these answers ourselves, drawing on our expertise in morphology. The information contained in Kaikki was verified using the FEW, particularly for the extraction of etymons. In cases where the context in Kaikki is incomplete, the expected response is supplemented with information from the FEW. Indeed, the goal of our study is not only to test the LLM's extraction capabilities, but also the relevance of the context. An incomplete context should be penalized. Thus, in the improvements to our methodology (see Chapter 5), we not only tested other models, but also found ways to expand the context of Kaikki when it was incomplete.

Chapter 4

First results and assessments

In the previous chapter, we have described the script we have written and that incorporates the findings from our experiments (no few-shot training, fixed output format, etc. see Section 3.2 for more details). Then, we have developed a reference dataset to evaluate the LLM's responses. Then, we ran the script on the 100 entries in the reference dataset to compare the LLM's responses with those in our gold dataset. The script presented in Appendix B¹ was designed so that each lemma is processed independently of the others: a new query is constructed for each entry, without retaining the history of previous responses. The responses generated by the model therefore do not influence its own future responses. The entries were nonetheless randomized before processing to avoid potential bias related to their order, particularly in the event that the protocol were to evolve.

This chapter presents an evaluation of the results obtained with the Mistral² model. We chose this model because it is easy to use. It does not cause compatibility errors between different libraries, unlike other models. This evaluation is conducted along two dimensions: a qualitative analysis and a quantitative analysis. The latter is performed using two scoring systems: manual annotation and automatic metrics.

4.1 Manual annotation

Manual annotation involved assigning scores and comments to each response from the LLM. Each question has its own guidelines (see presentation below in Section 4.1.1) to ensure independent evaluation. The goal of this annotation was to identify common errors, assess the model's linguistic relevance, and formulate general observations to facilitate analysis of the results. This provides a qualitative assessment of the model's outputs, which cannot be fully captured by automatic metrics. These observations can then be used to improve the model's performance.

4.1.1 Guidelines

The following summarizes, for each of the seven questions (morphological process, affix, stem, etymon/meaning pair, language, base, etymological derivational family), the scoring grid used as well as the decision rules adopted during the annotation process.

¹See for the complete version here: https://github.com/maiWenn217/Phononette/blob/main/src/mistral_prompt.py

²See <https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.3>

Procédé morphologique (proc) = morphological process

We evaluate whether the LLM correctly identifies the word-formation process(es) (suffixation and prefixation) based on the provided etymological context, according to the guidelines below. An example of each of these conditions is illustrated in Table 4.1.

- 1: the LLM provides the correct answer (row 1)
- 0.5:
 - the LLM does not respond, but there is no information about the morphological process in the context (row 2)
 - the LLM makes a mistake by extracting something from the context, even though that context does not contain information about the morphological process (row 3)
- 0:
 - the LLM makes a mistake even though the information is in the context (row 4)
 - the LLM does not provide an answer even though the information is explicitly stated in the context (it literally says there is a suffix or a prefix) (row 5) or implicitly (the word “suffix” or “prefix” is not written in the context, but the exact prefix or suffix is mentioned with a hyphen) (rows 6 and 7)

row	lemma	etym	gold dataset	LLM	score
1	adipeux	Dérivé savant du latin adeps, adipis (« graisse ») avec le suffixe -eux.	suffixation	Suffixation	1
2	bibition	bas latin bibitio	suffixation		0.5
3	antifébrile	Composé de anti-, fébrile et fièvre.	préfixation + suffixation	Composé	0.5
4	actinique	Composé du grec ancien ἀκτίς, ἀκτινός, aktis, aktinos (« rayon ») et du suffixe -ique.	suffixation	Composé	0
5	fabisme	Du latin faba (« fève ») avec le suffixe -isme.	suffixation		0
6	céphalaire	Du grec ancien κεφαλή, képhalê (« tête ») et -aire.	suffixation		0
7	crustacé	Du latin crusta (« croûte ») et -acé.	suffixation		0

TABLE 4.1: Examples of ratings for proc

These guidelines clearly state that we penalize the context. If the LLM did not find the correct answer, then we can’t give it a 1, but we also can’t give it a 0 because it followed the prompt’s instruction, which was: “If you can’t find the information, then don’t answer”.

Affixe (aff) = affix

We evaluate whether the LLM correctly identifies the affix that forms the lexeme, based on the guidelines below. An example of each of these conditions is shown in Table 4.2.

- 1: the LLM provides the correct answer (rows 1, 2 and 3)
- 0.5: the LLM provides an answer that is very close to what we expect (row 4)
- 0: the LLM is wrong; its answer has nothing to do with what we expect (rows 5 and 6)

row	lemma	etym	gold dataset	LLM	score
1	céphalaire	Du grec ancien κεφαλή, képhalê (« tête ») et -aire.	-aire	-aire	1
2	crustacé	Du latin crusta (« croûte ») et -acé.	-acé	-acé	1
3	actinique	Composé du grec ancien ἄκτις, ἁκτῖνος, aktis, aktinos (« rayon ») et du suffixe -ique.	-ique	-ique	1
4	carnassier	De l’occitan carnassier.	-ier	-ièr	0.5
5	chthonien	Du grec ancien χθόνιος, khthónios (« souterrain »), de χθών, khthón (« terre »).	-ien	khthónios	0
6	coturnisme	Du latin coturnix « caille » .	-isme	coturn	0

TABLE 4.2: Examples of ratings for aff

It should be noted that if the LLM had answered *aire* (without a hyphen) rather than *-aire* (with a hyphen) for CÉPHALAIRE (row 1), we would have accepted its answer and assigned a score of 1.

Radical (rad) = stem

We evaluate whether the LLM correctly identifies the stem (radical in French) of the lexeme (the part that remains after affixes are removed) according to the guidelines below. An example of each of these conditions is shown in Table 4.3.

- 1: the LLM provides the correct answer (row 1)
- 0.5: the LLM provides an answer that is very close to what we expect, moreover, it is consistent, that is, the segmentation is consistent, and the sequence it provides matches the input (row 2)
- 0:
 - the LLM is wrong; its response does not match what we expect (row 3)
 - the segmentation of the response is considered irrelevant (row 4)
 - the suggested sequence cannot be found in the entry (row 5)

This time, we do not penalize the context, since the information about the stem is never explicitly provided. It is information that can only be inferred based on the affix that is omitted from the entry.

row	lemma	gold dataset	LLM	score
1	coturnisme	coturn-	coturn	1
2	crustacé	crust-	crusta	0.5
3	sigillaire	sigill-	signum	0
4	viaire	vi-	via	0
5	actinique	actin-	actis	0

TABLE 4.3: Examples of ratings for rad

The segmentation of CRUSTACÉ (row 2) is consistent because the LLM decided to segment based on syllables. We consider that this segmentation makes sense, unlike the one proposed for VIAIRE, which does not seem natural because it does not segment at the syllable level. Furthermore, the segmentation occurs in the middle of a phoneme: indeed, if we take the phonological form of VIAIRE, which is /vjɛr/, we see that the phoneme ϵ corresponds to the spelling “ai”, so it would make no sense to segment between “a” and “i”.

Couple étymon / sens (etym_mean) = etymon-meaning pair

We evaluate a list of words pairs (etymology, meaning) provided by the LLM to trace the origin and meaning of the lexeme according to the guidelines below. An example of each of these conditions is shown in Table 4.4.

- 1: the LLM provides the correct answer (rows 1, 2 and 3)
- 0.5:
 - when we expect multiple etymons, the LLM provides only one that is correct and turns out to be the most important one to extract (row 4)
 - the LLM does not provide an answer, and there is no etymon in the context (row 5)
 - when we are expecting only one etymon, the LLM gives two, one of which is the correct answer, and the other has nothing to do with it (row 6)
- 0:
 - the LLM returns something that is not an etymon, even though the meaning is correct (row 7)
 - the LLM does not respond at all, even though there is an etymon to be extracted from the context (row 8)

It should be noted that variations are accepted. The LLM’s response will be considered perfectly correct, even if it has added or removed diacritics. When the expected etymon is a verb, as with POTABLE in row 2, the verb may be given in various forms: the infinitive (potare), the supine (potatum), the perfect (potus), etc. Definitions may also contain variations. For example, for IGNÉ in row 3, the expected definition for its etymon igneus is ‘de feu’. The LLM responded ‘Qui est de feu, qui a les qualités du feu.’³, and still received a score of 1.

³In this specific case, the LLM extracted the definition provided in the context rather than the one given in the etymology.

row	lemma	etym	gold dataset	LLM	score
1	thélite	Du grec ancien $\theta\epsilon\lambda\acute{\epsilon}$, thêlê (« mamelle ») et -ite.	thêlê, 'mamelle'	thêlê, 'mamelle'	1
2	potable	Du latin potabilis (« qui peut être bu, potable »), dérivé de potare (« boire »). Fréquentatif de bibo (« boire ») fait sur son parfait potus.	potabilis, 'qui peut être bu' potare, 'boire'	potabilis, 'qui peut être bu' potus, 'boire'	1
3	igné	Du latin igneus (« de feu »). Dérivé de ignis (« feu »), avec le suffixe -eus.	igneus, 'de feu' ignis, 'feu'	igneus, 'Qui est de feu, qui a les qualités du feu.' ignis, 'feu'	1
4	chthonien	Du grec ancien $\chi\theta\acute{o}\nu\iota\omicron\varsigma$, khthónios (« souterrain »), de $\chi\theta\acute{\omega}\nu$, khthón (« terre »).	khthónios, 'souterrain' khthón, 'terre'	khthón, 'terre'	0.5
5	ovoïde	Mot dérivé de ové, avec le suffixe -oïde.	ovum, 'œuf'		0.5
6	kérosène	Emprunté à l'anglais kerosene, lui-même dérivé du grec $\kappa\epsilon\rho\acute{o}\varsigma$ kêros, « cire » avec le suffixe anglais -ene (français -ène).	kêros, 'cire'	kêros, 'cire' anglais, 'anglais'	0.5
7	caprin	Du latin caprinus (« de chèvre »), de caper (« bouc ») et capra (« chèvre »).	caprinus, 'de chèvre' capra, 'chèvre'	latin, 'de chèvre' latin, 'bouc' latin, 'chèvre'	0
8	caséeux	Du latin caseus, « fromage » .	caseus, 'fromage'		0

TABLE 4.4: Examples of ratings for etym_mean

Langue source (lang) = source language

We evaluate the source language specified by the LLM for the direct etymon of the lexeme according to the guidelines below. An example of each of these conditions is shown in Table 4.5.

- 1: the LLM provides the correct answer (row 1)
- 0.5: the LLM provides a relevant answer, but it is not entirely correct (row 2)
- 0: the LLM is wrong (row 3)

As we can see with POLYADELPHE (row 2), we decided that when the LLM responds with “grec” instead of “grec ancien”, we should penalize its response. Indeed, Ancient Greek and (modern) Greek are two distinct languages just like Old French and

row	lemma	etym	gold dataset	LLM	score
1	visible	Du latin visibilis.	latin	latin	1
2	polyadelphie	Mot dérivé de adelphe (« frère »), avec le préfixe poly- (« plusieurs »).	grec ancien	grec	0.5
3	studieux	Du latin studiosus.	latin	italien	0

TABLE 4.5: Examples of ratings for lang

(modern) French. On the other hand, we did not make a distinction between “bas latin”⁴ and “latin”, since Vulgar Latin is a variant⁵ of Latin.

Base sémantique (base) = semantic base

We evaluate whether the LLM correctly captures the meaning (the semantic “base”) of the stem, according to the guidelines below. An example of each of these conditions is shown in Table 4.6.

- 1: the LLM provides the correct answer (row 1)
- 0.5: the LLM provides an answer that is close to what we expect (row 2)
- 0 :
 - the LLM provides an answer that has nothing to do with what we expect (it responds the stem (row 3), the etymon (row 4) or the entry itself (row 5)), even though the information is in the context
 - the LLM does not respond, even though the information is in the context (row 6)

row	lemma	etym	gold dataset	LLM	score
1	céleste	Du latin caelestis, dérivé de caelum (ciel).	ciel	ciel	1
2	oblation	Du latin oblatio (« offrande »).	offrir	offrande	0.5
3	adipeux	Dérivé savant du latin adeps, adipis (« graisse ») avec le suffixe -eux.	graisse	adip-	0
4	caprin	Du latin caprinus (« de chèvre »), de caper (« bouc ») et capra (« chèvre »).	chèvre	capra	0
5	cardiaque	Du latin cardiacus.	cœur	cardiaque	0
6	marmoréen	Du latin marmoreus (« de marbre »).	marbre		0

TABLE 4.6: Examples of ratings for base

⁴It corresponds to Vulgar Latin.

⁵More specifically, it is a historical subvariety of Latin, since it corresponds to the spoken Latin of the 3rd through 6th centuries.

Famille dérivationnelle étymologique (etym_fam) = etymological derivational family

We evaluate the list proposed by the LLM of lexemes that belong to the same derivational family as the entry and share the same stem, according to the guidelines below. An example of each of these conditions is shown in Table 4.7.

- 1: the LLM provides the correct answer (row 1)
- 0.5:
 - the LLM returns all the expected lexemes, but the entry itself is included (row 2)
 - the LLM provides a generally good answer with a minor error (row 3)
- 0:
 - the LLM generates too many lexemes that do not match what we want (row 4)
 - the LLM provides an irrelevant response, even though we are not expecting one (row 5)

row	lemma	gold dataset	LLM	score
1	oblation	oblat, oblatif, oblativité	oblat, oblatif, oblativité	1
2	caséeux	caséiforme, caséine, caséum	caséiforme, caséine, caséum, caséeux	0.5
3	thélite	thélalgie, thélorrhagie, thélotisme	thélalgie, thélorrhagie, thélotisme, mammelon	0.5
4	potable	potablement, potabilité, impotable	potablement, comestible , bibition , boire , embuvage , imbuvable , impotable	0
5	thalassique		maritime	0

TABLE 4.7: Examples of ratings for etym_fam

4.1.2 Qualitative analysis

Manual annotation revealed several limitations of the current prompt, as well as some unexpected behaviors of the model. First, we observed very encouraging results regarding the issue of source language: the model performed excellently. Nearly all the responses were correct (37 out of 50 suppletive entries), with only a slight difficulty distinguishing Greek from Ancient Greek. However, we observed that the model did not react as we had expected with regard to non-derived entries. In fact, when compiling the reference dataset, it was decided that non-derived lexemes should receive no response (see Table 4.8). The idea was to ask it about the derivational process first so that it could determine whether or not the lexeme was

lemma	etym	proc	aff	rad	etym_mean	lang	base	etym_fam
oblation	Du latin oblatio (« offrande »).	suffixation	-ion	oblat-	oblatio, 'offrande'	latin	offrir	oblationnaire, oblat, oblatif
phrase	Du latin phrasis.	NaN	NaN	NaN	NaN	NaN	NaN	NaN

TABLE 4.8: What we expect for a non-derived entry

derived. We expected that, if the LLM could not identify any process, it would categorize the input as non-derived and not respond to the subsequent questions (as explicitly stated in the prompt). Indeed, if the lexeme is not derived synchronically, identifying its current affix, stem, or base would be meaningless, hence our expectation of no response. However, this expectation proved to be too strict for cases, such as *FRIABLE*, where a simple word in modern French nevertheless has an etymological description mentioning an ancient formation process (see Section 3.3). However, this does not explain why the LLM did not follow the instruction. In fact, this type of edge case, such as *FRIABLE*, is rather rare because among the 50 non-derived lexemes we selected, none of these lexemes mentioned an ancient affixal formation. Although the LLM had correctly identified the absence of a morphological process, it still provided an answer to the other questions (e.g., it suggests *phras-* as the stem of *PHRASE*).

However, failing to follow this instruction may have been beneficial. In fact, the model is sometimes able to correctly identify the affix of a derived lexeme, even though it had incorrectly categorized it as non-derived (6 suppletive entries out of 50). For example, it states that *CRUSTACÉ* is morphologically simple (Table 4.1, row 7) but suffixed with *-acé* (Table 4.2, row 2). We did not expect that the model would be able to extract the affix but not the morphological process. This means that this instruction could lead to the exclusion of certain relevant responses. The problem likely stems from the fact that we did not clearly define what we meant by morphological process and affix. One possible approach would be to explicitly limit the set of processes considered in the prompt (prefixation and suffixation). It might therefore be appropriate to instruct the model that only these affixation processes, as well as the absence of derivation, are possible answers. Next, we could explain precisely what prefixation entails and what suffixation entails. Particularly in cases where the information is not explicit in the context, as with *LACTIQUE* (8), but is implied, as with *THÉLITE* (9). Indeed, in the latter, it is not clearly stated that this is a suffixation, but the suffix *-ite*, which corresponds to the end of the entry, suggests that we are dealing with a suffixation. Once these concepts have been made explicit to the LLM, it may have less difficulty extracting the derivational process.

(8) *LACTIQUE* → Du latin *lac, lactis* (« lait ») avec le **suffixe** *-ique*.

(9) *THÉLITE* → Du grec ancien *θηλή*, *thêlē* (« mamelle ») et **-ite**.

The analysis revealed further inconsistencies regarding the morphological knowledge used by the model. Sometimes, it identifies the affix but not the stem, and *vice versa* (14 suppletive entries out of 50). For example, for *ACTINIQUE*, it identifies the suffix *-ique* (Table 4.2, row 3) but not the stem *actin-* (Table 4.3, row 5), and for *COTURNISME*, it identifies the stem *coturn-* (Table 4.3, row 1) but not the suffix *-isme* (Table 4.2, row 6). However, identifying one should normally make it easier to identify the other, since these pieces of information are closely related: the stem is the part that remains when we remove the affix, and the affix is the part we add to the stem. It might therefore be useful to further clarify these relationships in the prompt. We can nevertheless note that, when it provides the stem or affix, the model places the hyphen in the correct position (17 suppletive entries out of 50), as was the case for *LIGNEUX* (10), which provides no indication of its suffix or its stem, but the LLM still managed to extract its stem as follows: “*lign-*”. This detail suggests that it possesses a relatively accurate representation of these morphological concepts, even if this knowledge remains imperfect.

(10) *LIGNEUX* → Du latin *lignosus* (« comme du bois »), de *lignum* (« bois »).

Clearly defining each linguistic concept seems to be one of the most important improvements needed to prevent the model from providing inconsistent answers. This would, for example, prevent the model from responding with the source language when asked for the etymon.

We encounter this same problem of confusion with the base: instead of the base, the LLM returns the stem, the etymon, or even the entry itself. For example, for ADIPEUX (Table 4.6, row 3), it returns adip- (stem) instead of ‘graisse’, for CAPRIN (Table 4.6, row 4), it returns capra (etymon) instead of ‘chèvre’ and for CARDIAQUE (Table 4.6, row 5), it responds with “cardiaque” (entry) instead of ‘cœur’. It would likely be necessary to add one or more examples (few-shot) to resolve all these ambiguities.

Overall, the LLM followed the instructions we gave it fairly well, among other things, when information was missing, it did not try to make it up in most cases, except when the linguistic concept was not understood. However, it performed disastrously when we asked it to extract lexemes from the same family as the entry that shared the same etymological stem. In fact, for all entries, it simply copied all the information present in the context rather than selecting only the relevant members. This problem may be related to the difficulties encountered previously in identifying the stem. Once the model is better able to identify the stem, we can expect it to select the desired lexemes more effectively. In fact, we can guide it by adding to the prompt that anything to be selected must contain the sequence of the identified stem.

One final technical observation concerns the maximum limit on the number of tokens (the *max tokens* parameter indicates the maximum number of words the LLM can include in its response). This value proved to be insufficient for certain inputs, resulting in incomplete responses or even no response at all. An increase in this limit therefore appears necessary to allow the model to provide all the expected information.

Observation	Solution
answers for non-derivatives	define the proc and aff notions
morphological inconsistencies proc/aff	
morphological inconsistencies aff/rad	explain the relation between an affix (what we add to the stem) and a stem (what remains after removing the affix it)
confusion etym_mean with lang	add examples of what we expect (few-shot)
confusion base with rad, etymon, entry	
catastrophic for etym_fam	solve the issue with rad then, ask him to extract all the words that contains this stem
excellent for lang	
limit <i>max tokens</i> set too low	increase the limit

TABLE 4.9: Summary of the qualitative analysis

All of these observations and proposed solutions are summarized in Table 4.9. Overall, one solution that comes up consistently, regardless of the question, is to define the key concepts. Another solution would be to change our approach by querying

the LLM with a different prompt for each question and incorporating its response into the context for subsequent questions asked about the same input. This would allow for more detailed prompts without overwhelming the LLM with an excessively long prompt: we can thus be more specific while avoiding the lost in the middle effect.

4.2 Automated metrics

During manual annotation, we simultaneously evaluated two aspects: first, the completeness of the context (that is, its ability to contain all the information necessary to answer the question) and, second, the quality of the information extracted by the LLM. The goal is to improve these two components as much as possible, until we achieve performance levels deemed satisfactory. Ideally, the answers produced by the model should match exactly the reference (gold) answers. In order to effectively compare different prompts and different models, the experiments are conducted solely on the inputs present in our reference dataset. Once the first annotation phase is complete, it is no longer feasible to continue with a fully manual evaluation for each experiment, as this would be far too time-consuming. Subsequent evaluations will therefore rely on the heuristic metrics developed from this initial annotation. These metrics are calibrated against the manually annotated reference dataset and are therefore intended to replicate human judgment as closely as possible. It is therefore expected that they cannot be directly applied on a large scale in the absence of a reference dataset. To evaluate new data, it will be necessary to assemble a representative sample, manually determine the ground-truth answers for this sample, and then use these same metrics to estimate the quality of the answers produced by the LLM. To verify whether the scores provided by the metrics are indeed identical to, or at least very close to, human scores, we use the statistical measures and tests presented below.

RMSE The Root Mean Squared Error (Chai and Draxler, 2014) is a measure of the distance between two paired lists of values. For two lists of scores $l = [s_1, \dots, s_n]$ and $l' = [s'_1, \dots, s'_n]$ of the same size n , the RMSE between the two is calculated as follows:

$$\text{RMSE}(l, l') = \sqrt{\frac{1}{n} \sum_{i=1}^n (s_i - s'_i)^2} \quad (11)$$

It corresponds to the Euclidean distance (i.e., the standard distance between two points in a space), multiplied by a normalization factor $\sqrt{\frac{1}{n}}$. This measure provides an indication of the distance between the manual annotation and the metrics: it is an indicator of the LLM's error rate. It should not be interpreted in isolation but compared with the RMSE of the other questions. Thanks to this metric, we can determine on which question the LLM made the most errors. If we have an RMSE of 0.8 for the affix question and an RMSE of 2 for the base question, then we can say that the metrics were more often incorrect for the base question than for the affix question.

Cohen's Kappa The Cohen's Kappa (Cohen, 1960) is a measure of Inter-Annotator Agreement (IAA). In this study, it is used to evaluate the degree of agreement between human scores and metric scores. It is calculated as follows:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (12)$$

where p_o is the observed proportion of agreement, and p_e is the expected proportion of agreement. The κ score is necessarily less than or equal to 1 and may be negative. If $\kappa = 1$, then the agreement is perfect: the two sets of scores are identical. If $\kappa < 0$, the observed agreement is less than the expected agreement: we then consider that the annotators are in disagreement. $\kappa > 0$ is not sufficient to demonstrate a reasonable degree of agreement among annotators, and the value for a "good κ " varies depending on the annotation tasks. In our case, we will consider that a κ greater than 0.7 indicates a desirable degree of agreement between our scores and those of the metrics.

Pearson's r Pearson's r (Pearson, 1895) is a measure of correlation between two paired data sets $l = [s_1, \dots, s_n]$ and $l' = [s'_1, \dots, s'_n]$. It is given by the following formula:

$$r = \frac{\sum_{i=1}^n (s_i - m)(s'_i - m')}{\sqrt{\sum_{i=1}^n (s_i - m)^2 \sum_{i=1}^n (s'_i - m')^2}} \quad (13)$$

where m is the average of the values of l and m' is the average of the values of l' . The r thus obtained is a value between -1 and 1. If $r = 1$, the two sets of scores are positively correlated, that is, to put it simply, they are proportional (e.g., $l = [1, 2, 3]$ et $l' = [2, 4, 6]$). If $r = -1$, the two sets of scores are negatively correlated, that is, to put it simply, inversely proportional (e.g., $l = [2, 4, 6]$ et $l' = [6, 4, 2]$). An absence of correlation between the two is indicated by a value of $r = 0$.

T-test The (paired) t-test (Student, 1908) is a statistical test used to determine whether two paired sets of data $l = [s_1, \dots, s_n]$ and $l' = [s'_1, \dots, s'_n]$ have the same expectation value (i.e., the expectation value is the expected mean value, it is a kind of theoretical mean). The test calculates a value α , called the p-value, which corresponds to the probability of obtaining such values given that the two data sets have the same expectation. When α is below a certain threshold (generally set at 0.05), then we take a very small risk in asserting that l and l' do not have the same expectation value. When α is above the threshold, then the risk is too great: we cannot reject the hypothesis of equal expectation values. Having a high p-value is not sufficient to conclude that the two datasets have the same expectation (which would indicate that the two distributions are close), but it provides a good intuition on the matter. We use this test in conjunction with *Pearson's r*. Indeed, the latter indicates correlation but not the position of the values (e.g., $l = [1, 2, 3]$ and $l = [4, 5, 6]$ are correlated but far apart). With the t-test, we ensure that it is impossible to claim that the theoretical mean of l and that of l' do not match. Thus, we would expect a p-value greater than the threshold of 0.05.

We will now explain, for each question, how the metrics were developed to most closely approximate human scores. The results of the four metrics (RMSE, *Cohen's*

Kappa, *Pearson's r*, *t*-test) for each question are presented in Appendix D (Table D.1).

Proc

To follow the guidelines established for the morphological process, we first checked whether information about the process could be found in the context. To determine whether the information is present explicitly, we simply check whether the sequences “suffixation”, “prefixation”, “suffix”, or “prefix” are present in the context. To determine whether it is present implicitly, we use regular expressions that search for a pattern that begins or ends with a hyphen. Based on this information, our metric assigns scores of 0, 0.5, and 1 according to the various conditions listed in Section 4.1.1.

The results of the measurements and tests revealed that the scores provided by our metric were very similar to the manual annotations. Indeed, the *Pearson's* correlation coefficient indicates a strong correlation, and the *Cohen's Kappa* shows a high degree of agreement between human annotations and the metric. Furthermore, since the *p*-value of the *t*-test is well above 0.05, we cannot conclude that there is a difference between the means.

Aff

As with the morphological process, we begin by checking whether the expected affix appears in the context provided to the LLM. When it does, a simple comparison between the LLM's response and the gold answer allows us to decide immediately. However, when the LLM proposes a response even though the affix does not appear explicitly in the context, a strict comparison is no longer sufficient to account for cases where the response, although incorrect, remains linguistically close to the correct answer (for example, *-ièr* for *-ier*, see Table 4.2, row 4). To address this, we used the Levenshtein distance (Levenshtein, 1965). This metric calculates the minimum number of operations (insertion, deletion, or substitution of a character) required to transform one string into another. Based on this distance, we calculate a proximity ratio, normalized between 0 and 1, using the following formula:

$$\text{ratio} = 1 - \frac{\text{Levenshtein}(\text{gold_answer}, \text{llm_answer})}{m} \quad (14)$$

where m is the maximum of the lengths of `gold_answer` (the expected answer) et `llm_answer` (the given answer).

This ratio produces continuous values, whereas human annotation relies on only three discrete values (0, 0.5, and 1). The calculation of *Cohen's Kappa* is not defined for continuous values, it is therefore necessary to convert this ratio into a discrete value comparable to that assigned by the human annotator. Since the goal of these metrics is to best approximate human judgment, we have chosen to divide the interval [0,1] into three equal parts, with each part rounded to the nearest discrete value:

$$\underbrace{0 \text{ --- } \frac{1}{3}}_{\text{arrondi à 0}} \quad \underbrace{\frac{1}{3} \text{ --- } \frac{2}{3}}_{\text{rounded to 0.5}} \quad \underbrace{\frac{2}{3} \text{ --- } 1}_{\text{rounded to 1}}$$

The *Pearson's* correlation coefficient once again shows that there is a strong correlation, and the annotators are in broad agreement. However, the *p*-value from the

t -test is slightly less than 0.05. This means that the null hypothesis can be rejected, we can conclude that there is a difference in the means.

To determine whether this difference in means reflects a real disparity between the two sets of scores, we compared their distributions. A figure in Appendix C (Figure C.1) shows the distributions of the human and automatic annotations in the form of a histogram. In this histogram, we observe that the great majority of responses receive a score of 0 in both cases, with slightly more responses scored as 0 by the human evaluator. A very small number of responses receive a score of 0.5, and this is evenly split between the two annotations. Finally, an even smaller proportion receives a score of 1, with slightly more responses scored as 1 by the metric this time. We can conclude that there is no marked discrepancy between the two distributions across the three scores. The difference in means detected by the t -test, although statistically significant, represents a gap that is too small to reflect a complete disagreement between the human annotations and the metric. We therefore conclude that the metric's scores are sufficiently similar to the human annotations.

Rad

To get closer to human scores by following the guidelines, we do not need to consider the context this time. However, we do need to split the inputs into syllables to determine whether the LLM provides a relevant response. To do this, we used the Pyphen library⁶. For each language, Pyphen relies on a dictionary of patterns that specifies the positions where a break is allowed within a word. These patterns are compared to the word being analyzed and then combined to determine the most likely syllabic segmentation positions. The resulting segmentation does not necessarily correspond to strict phonological syllabification, but rather to an orthographic segmentation appropriate for the language in question. In this project, this segmentation is used solely to verify that the stems proposed by the LLM adhere to plausible syllabic boundaries. For example, for the lexeme VIAIRE, Pyphen proposes the segmentation *viaire* → *vi-ai-re*. This information makes it possible to verify that a stem such as *vi-* corresponds to a plausible orthographic break, whereas a segmentation such as *via-ire* or *v-iai-re* does not.

Once again, our measurements clearly indicate a very strong similarity between the two sets of scores.

Etym_mean

To match human scores for pairs (etymon, meaning), we must evaluate two distinct elements: the validity of the proposed etymon and the validity of the meaning associated with it. As for the etymon, we compare its character string with that of the gold: if they are equal, then the etymon is valid. In accordance with the guidelines, we consider the forms of the etymon with and without diacritics to be equal. However, when it comes to the meaning, we cannot simply compare the character strings. In fact, to remain compliant with the guidelines, we accept different formulations (e.g., for IGNÉ, 'de feu' and 'Qui est de feu, qui a les qualités du feu.' are both considered correct). We therefore fed the meaning (whether a word or a phrase) into an *encoder*⁷ in order to measure the semantic similarity between the expected meaning and the meaning proposed by the LLM.

⁶See <https://github.com/Kozea/Pyphen>

⁷An *encoder* creates a numerical representation of the input, this representation can be a vector or an embedding, as opposed to a *decoder*, also known as a *language model*, which generates text.

The `biencoder-distilcamembert-mmarcoFR`⁸ that we used is a French bi-encoder based on CamemBERT, trained on semantic search tasks and therefore particularly well-suited for comparing short sequences of meanings in French. This encoder transforms the sequence into a vector in a vector space, with its position indicating its meaning. For example, we expect the vectors representing ‘de feu’ and ‘Qui est de feu, qui a les qualité de feu.’ to be close together in the vector space. This vectorization is dynamic, meaning that the vector corresponding to the word (or phrase) is calculated the moment the words (or phrases) are fed into it: these are not predefined vectors. Once this model assigns a vector to each of the two words (or phrases), we calculate the cosine similarity between the two. The cosine similarity between two vectors corresponds to the cosine of their angle: two vectors are therefore similar if the angle between them is small, that is, if their directions are close. We define two meanings as semantically equivalent when their cosine similarity exceeds a threshold empirically set at 0.6.

Trivial cases are handled separately: two strictly identical strings receive a similarity score of 1, and if only one of the two meanings is an empty string, the score is set to 0. Now we can determine whether an etymon is valid or whether a meaning is valid. But things get more complicated when multiple etymons are expected. We decide that if we find at least one valid pair in the LLM’s response, then the metric assigns a score of 0.5. Another case we consider with our metric is when the LLM does not respond but the expected information is not in the context, in this case, the metric assigns a score of 0.5. In fact, we check whether the expected etymons appear in the context.

The results obtained show a strong correlation between human scores and those of the metric ($r = 0.89$), with a *Cohen’s Kappa* of 0.77 indicating a level of agreement deemed satisfactory. However, this is the question for which agreement is lowest among the seven metrics developed, which is likely due to the difficulty of automating a judgment of semantic similarity, which is more subjective than the purely formal criteria used for *proc*, *aff*, or *rad*. The RMSE (0.20) is also the highest in the table, which supports this conclusion. The *t*-test, on the other hand, yields a *p*-value of 1.00, well above the threshold of 0.05: we therefore cannot reject the hypothesis that the two score distributions have the same expected value.

Lang

For this question, creating the metric was very easy, since the guidelines were extremely simple: 1 if it’s correct, 0 if it’s incorrect, and 0.5 for cases where the LLM responds with “grec” instead of “grec ancien”. Thus, our metric returns 0.5 when the LLM’s response is included in the gold answer, 1 when the responses match perfectly, and 0 otherwise. As with human annotation, the responses “latin” and “bas latin” are considered equivalent.

The results show that the human annotator and the metric annotated the data exactly the same way: all scores are identical ($\kappa = 1$).

Base

Regarding the semantic base, we must first and foremost ensure that the metric does not validate an answer that confuses the concept of the base with another morphological concept. Therefore, before comparing the answer to the gold standard, we filter out cases where the LLM’s response matches either the entry itself, a stem,

⁸See <https://huggingface.co/antoinelouis/biencoder-distilcamembert-mmarcoFR>

or an affix, identifiable by the presence of a hyphen at the beginning or end of the string. In these two cases, the metric directly assigns a score of 0, without even comparing it to the gold. However, the case where the LLM returns the etymon rather than the base is not detected by a dedicated rule: it is filtered out only if its similarity to the gold proves insufficient in the next step. Once these ambiguities have been filtered out, we proceed in the same way as for `etym_mean`: an exact match between the gold response and the LLM’s response yields a score of 1, and we reuse the `biencoder-distilcamembert-mmarcoFR` encoder to measure, using cosine similarity, whether the LLM’s response is close enough to the expected meaning without being identical to it (Table 4.6, row 2, where “offrir” is judged to be close to “offrande”). However, the similarity threshold was lowered to 0.5 (compared to 0.6 for `etym_mean`). Below this threshold, the metric assigns a score of 0.

The results obtained show a *Pearson’s* correlation coefficient of 0.97, as well as a *Cohen’s Kappa* of 0.90, indicating a level of agreement deemed satisfactory with the human annotation. The *t*-test yields a *p*-value of 0.57, which is well above 0.05, therefore, we cannot reject the hypothesis that the expected values of the human scores and the metric’s scores are the same.

Etym_fam

The question of etymological derivational families involves comparing two lists of lexemes rather than two single values, which led us to use the Jaccard index (Jaccard, 1901). This measure indicates the extent to which two sets A and B overlap, according to the following formula:

$$\text{Jaccard}(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (15)$$

where $A \cap B$ denotes the intersection of the sets, that is, the set of their common elements, $A \cup B$ denotes the union of the sets, that is, the set of elements belonging to one or the other, and $|X|$ denotes the cardinality of the set X , that is, the number of its elements. Using the Jaccard index in our case amounts to measuring how many elements our two lists have in common, relative to their sizes. The Jaccard index is 1 when the two lists are identical (or empty), and 0 when they have no elements in common. A Jaccard index of 0.9 or higher between the two lists indicates a nearly perfect match and yields a score of 1, provided that the entry itself is not among the words suggested by the LLM (Table 4.7, row 1). Conversely, an index less than or equal to 0.4 means that the two lists have almost nothing in common and yields a score of 0. Between these two thresholds, we recalculate the Jaccard index by adding the entry itself to the gold list. If the index reaches at least 0.8, this means that the LLM has provided virtually all of the expected words, simply adding the entry itself results in a metric score of 0.5, in accordance with the guidelines (Table 4.7, row 2). Otherwise, the metric assigns a score of 0.

The results obtained show a *Pearson’s* correlation coefficient of 0.89. *Cohen’s Kappa*, on the other hand, is 0.69, the lowest value among the seven metrics developed and slightly below the set threshold of 0.7, which indicates a less clear agreement with human annotation than for the other questions. This is likely due to the fact that the Jaccard index, while effectively capturing the overall overlap between two lists, does not always distinguish the qualitative nuances that human annotators take into account (nature of the error, relevance of added or missing words). Nevertheless, the *t*-test yields a *p*-value of 0.08, which is above the 0.05 threshold: we

therefore cannot reject the hypothesis that the two score distributions have the same expected value, which remains encouraging despite the κ falling slightly below the desired threshold.

In conclusion, the results show that the metrics developed for each of the seven questions accurately replicate human judgment. For five of the seven questions (proc, rad, lang, base, and etym_mean), *Cohen's Kappa* exceeds 0.75 and *Pearson's r* is greater than 0.88, while the *t*-tests confirm that the score distributions of the human annotator and the metric are statistically indistinguishable. Only aff and etym_fam show slightly weaker agreement ($\kappa = 0.84$ and $\kappa = 0.69$, respectively), but even in these two cases, the discrepancy is due to marginal and easily identifiable cases (a difference of a single character in a comparison based on the Levenshtein distance, or the inclusion of the entry itself in a list comparison based on the Jaccard index), rather than systematic disagreement between the two annotation methods.

It is this overall level of agreement that allows us to scale up from the reference dataset of 100 entries to a much larger scale. The effort invested in this calibration step will allow us, in the following chapters, to compare prompts, contexts, and models (Chapter 5) while ensuring that the scores obtained accurately reflect the actual quality of the LLM's responses.

Chapter 5

Improve the results

Analysis of the first results has allowed us to identify the limitations of our current method. To improve these results, we can adjust three different variables: the prompt, the context, and the model. In this chapter, we will focus on the latter two, beginning by extending the etymological context of each entry and then testing different types of models.

5.1 Etymological extension

In our methodology, we evaluate both the relevance of the LLM and the completeness of the context. Following our analyses (qualitative and quantitative), we realized that a huge number of entries were incomplete. For example, regarding affixes, 31 of the 50 derived entries lack any contextual information. Furthermore, as mentioned in Section 3.2, the affix is key information for identifying the stem. One reason why the etymological description is incomplete is that Wiktionaire contributors seem to follow a recurring pattern for morphologically complex loanwords: “de [*ancient language*] [*already constructed etymon*] [*(meaning)*]”. Crucially, this pattern reflects a genuine absence of information to report: for these words, the morphological formation process did not take place in French but in the source language (e.g., Latin), the word was already fully constructed at the time of borrowing. Contributors therefore correctly document the etymology (the borrowing itself) without providing a morphological analysis. This is notably the case for NUCLÉÉ (16), which do not mention any formation process.

(16) NUCLÉÉ → Du latin nucleatus (« à noyau »).

Similarly, the Wiktionaire contributors seem to follow a different pattern when the lexeme is derived directly from an ancient language. In such cases, they provide the direct etymon and the process by which the French lexeme was formed. This is the case with ADIPEUX (17).

(17) ADIPEUX → Dérivé savant du latin adeps, adipis (« graisse ») avec le suffixe -eux.

It is therefore clear that a borrowed, complex lexeme is more likely to have an incomplete etymological description, and conversely, a lexeme directly derived from a simple Latin or Ancient Greek lexeme is more likely to have a rich etymological description. The idea is therefore simple: we will supplement the etymological description of the entry with those of all its etymons, all the way back to the base etymon. The goal is to increase our chances of obtaining an etymological description that includes information about the derivational process or the affix, and thus, to identify the stem. Extending the etymology would not only increase our chances of

obtaining complete descriptions of the derivational process or the affix, but also better document the historical evolution of the lexeme (etym_mean) and the meaning of the stem (base).

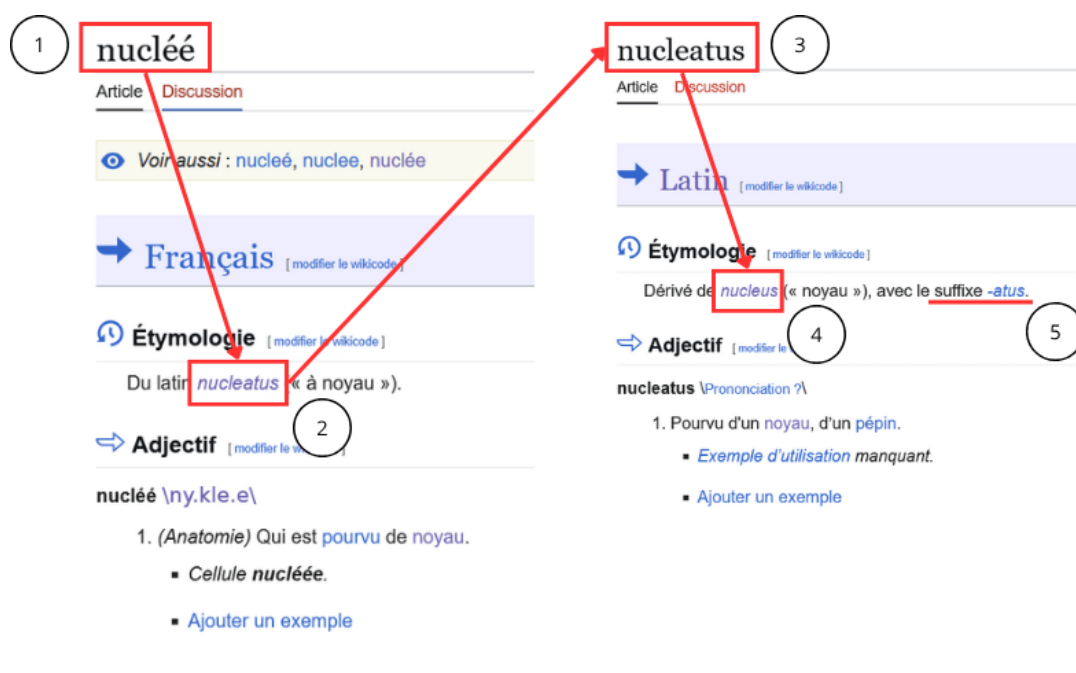


FIGURE 5.1: Tracking the etymological history of NUCLÉE into Wiktionnaire

In principle, tracing back to the base etymon in Wiktionnaire involves going to the page for the derived lexeme (step 1 in Figure 5.1) and clicking on the etymon whose base we want to know (step 2). By clicking on this etymon, we are taken to the dedicated Wiktionnaire page (step 3), which, under the “Étymologie” section, provides an etymological description that clearly states which etymon it derives from (step 4) and how it was formed (step 5). As we can see, when we trace the etymological history of the lexeme, we move from the “Français” portal to the “Latin” portal. It is therefore not surprising that the etymological description of the Latin etymon does not explicitly mention the suffix expected for the French lexeme. In fact, the description indicates that *nucleatus* is morphologically formed from *nucleus* using the suffix *-atus*. It does not state that *NUCLÉE* comes from *nucleatus* with the suffix *-é*. However, this information is sufficient. It is sufficient for the LLM to identify the suffix mentioned in the description (*-atus*), remove it from *nucleatus* to obtain the stem *nucle*, and then deduce, by comparison with the French word *NUCLÉE*, that the affix sought is indeed *-é* (since *nuclée* - *nucle* = *-é*).

To automatically extend the etymological context and navigate from page to page, as shown in Figure 5.1, we used Kaikki’s API (Application Programming Interface) to retrieve the pages for the desired entries. For example, to directly download the JSON file corresponding to *OBLATION*, we use the following URL:

```
https://kaikki.org/frwiktionary/All20languages20combined/meaning/<w>
/<wo>/<word>.jsonl
```

This URL can easily be adapted to access any word, provided it is present in Kaikki. Now, to access the JSON file for the most distant etymon and add its etymological description to the context, we must first identify that etymon. To do this, we use regular expressions based on three different patterns that we have identified.

Pattern 1 The first pattern follows the structure found in the etymological descriptions in (18a) and (18b). It accounts for 86% of the supplementary entries (43 out of 50), and here is its regular expression:

Pattern 1

`(de | du | de la | de l' | de l')\s* language \s+(\w+)`

a partitive article + a language + what is supposed to correspond to an etymon

- (18) a. CARNASSIER → De l'occitan carnassier.
b. FABISME → Du latin faba (« fève ») avec le suffixe -isme.

The following two patterns are intended to cover as many possible cases as possible, while acknowledging that complete coverage is likely unattainable.

Pattern 2 The second pattern, which we find in particular in the etymological description of FUNICULAIRE (19), accounts for 8% of the suppletive entries (4 out of 50). Here is its regular expression:

Pattern 2

`dérivé\w* de (\w+)`

the sequence *dérivé* followed by something (or nothing) that could resemble a language + the sequence *de* + what is supposed to correspond to an etymon

- (19) FUNICULAIRE → Dérivé de funicule, avec le suffixe -aire.

Pattern 3 The third pattern, which we find in particular in the etymological description of ANTIFÉBRILE (20), accounts for 2% of the suppletive entries (1 entry out of 50). Here is its regular expression:

Pattern 3

`^\w+ de (\w+)`

the etymological description begins with a character sequence (such as *Composé*) followed by the sequence *de* + what is supposed to correspond to an etymon

- (20) ANTIFÉBRILE → Composé de anti-, fébrile et fièvre.

Even though this pattern is not very common in our dataset, we expect it to be relatively frequent on a large scale. Finally, in Table 5.1, we have listed three examples that benefited from additional information after the etymology was extended. For the lexeme CAPRIN (row 1), we obtain information regarding the type of morphological process (suffixation), which provides an implicit clue about the French suffix and the stem. Indeed, according to the etymology, *caprinus* = *capr* + *inus*, which, when transposed into French, gives us the stem *capr* and the suffix *-in*. For the lexeme STUDIEUX (row 2), we obtain information about its base (*étude*), and for the lexeme FUNICULAIRE, we now know its base (*corde*) as well as the etymons constituting its direct etymology (*funiculus* and *funis*).

row	lemma	etym	etym_ext
1	caprin	Du latin <i>caprinus</i> (« de chèvre »), de <i>caper</i> (« bouc ») et <i>capra</i> (« chèvre »).	Dérivé de <i>caper</i> (« bouc, chèvre »), avec le suffixe <i>-inus</i> , de l'indo-européen commun <i>*kapro-</i> (« animal mâle »), qui donne le grec ancien <i>κάπρος</i> , <i>kápros</i> (« sanglier »), le sanscrit <i>kaprtha</i> (« pénis »), <i>gafr</i> en gallois, <i>gavr</i> en breton.
2	studieux	Du latin <i>studiosus</i> .	Dérivé de <i>studium</i> (« zèle, ardeur, soin étude »), avec le suffixe <i>-osus</i> , dérivé de <i>studeo</i> (« s'appliquer à, s'occuper avec zèle de »), avec le suffixe <i>-ium</i> , avec le préfixe <i>s-</i> , de <i>tundo</i> (« frapper »), le sens a glissé de « se lancer, se jeter sur » vers « s'appliquer, se dédier à » enfin « étudier » , apparenté au grec <i>σπεύδω</i> , <i>speúdō</i> (« se hâter, être diligent »).
3	funiculaire	Dérivé de <i>funicule</i> , avec le suffixe <i>-aire</i> .	Du latin <i>funiculus</i> , dérivé de <i>funis</i> (« corde »), avec le suffixe <i>-culus</i> , de l'indo-européen commun <i>*dhu-</i> (« corde »), qui donne <i>θῶμιγξ</i> , <i>thōminx</i> (« corde ») en grec ancien.

TABLE 5.1: Examples of entries with the extended etymology

5.2 Comparing of LLMs

We want to test our new context using extended etymology. Now that we have calibrated metrics, we can run tests on different models. In addition to the Mistral model we had already used, we selected three standard LLMs (Qwen¹, Gemma², and Llama³) and a thinking LLM (Deepseek⁴, based on Llama). All models are similar in size (between 7billions parameters and 9billions parameters) and were released between 2024 and 2025. Indeed, for comparison purposes, it is preferable that the models be similar in terms of size and year of release.

The Gemma, Llama, and Deepseek models were inconclusive. They failed to answer the questions in the correct format, and their responses were therefore rejected. This means that the current prompt is not suitable for these models, as it does not allow them to produce actionable responses. For the other models, the results are shown in Appendix D (Table D.2).

We observe that performance deteriorates with the new context. Indeed, Mistral achieves a lower score on all questions. We suspect that the context contains too much information. Upon closer inspection, it became apparent that our method for extending the etymology often went back too far, exceeding the scope of the direct etymology and drowning out relevant contextual information in noise. Furthermore, the prompt was not adapted to the new context.

Qwen appears to be the model with the best results (average=0.30), outperforming Mistral even when it relies on the original context.

¹See <https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>

²See <https://huggingface.co/google/gemma-2-9b>

³See <https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>

⁴See <https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B>

Chapter 6

Discussion and conclusion

6.1 Discussion

6.1.1 Kaikki: easily usable source

Among the existing resources, several are difficult to process automatically. This is particularly true of the TLFi. The TLF (Trésor de la Langue Française) is a 16-volume dictionary covering the 19th and 20th centuries; the TLFi is its electronic version. However, the information in the TLFi is accessible *via* its website (HTML) and not in the form of a tabular file (e.g., CSV or JSON), which complicates its use. In fact, the data must first be extracted, which is no easy task when the pages contain numerous nested tags, formatting elements, links, abbreviations, or special characters. A change to the website's structure can also render the extraction script obsolete. Then, once the data has been extracted, it must be cleaned and restructured before it can be analyzed or integrated into a computer processing system.

Ideally, we would have been able to use information from an etymological dictionary to ensure data quality. A relevant candidate for our project, which focuses on the French language and the origin of its words, is the FEW. Unfortunately, the online FEW is available only in image format and is therefore unusable.

Due to these constraints, we chose to work with Kaikki (and, therefore, Wiktionnaire). To verify the quality of the data, we manually checked the etymological section of 50 suppletive entries. For only 2 lexemes, this section was not complete enough to allow us to verify the origin of the stem. For example, for GRAVIDISME, it made no mention of an etymon :

(21) GRAVIDISME → Dérivé de gravide, avec le suffixe -isme.

For 46 lexemes, the information was correct, that is, consistent with that found in the FEW or other reference dictionaries (Littré, 1873-1874; Baumgartner and Ménard, 1996). This means that for only 2 lexemes, the etymon given in the etymological description does not match what is found in the FEW. Admittedly, Wiktionnaire does not have a great reputation, but human verification has clearly shown that it is a resource we can rely on to a large extent.

6.1.2 Marginal cases of conversion

Based on the definition in Tribout (2010), conversion occurs when there is a change in category between the base and the derivative, and when one of the stems in the stem-space of the base is identical to one of the stems in the stem-space of the derivative.

The pair CONCEVOIR > CONCEPT therefore constitutes a case of conversion even though the two stems are not identical (/kõsəvwar/) ≠ /kõsept/). Indeed, CONCEPT

has a semantic relation with CONCEVOIR¹ but it does not have a derivational exponent. /kɔ̃sɛpt/ corresponds to a learned stem of the verb CONCEVOIR (also found in CONCEPTEUR): the thirteenth stem, specific to verbs, used not in inflection but in the formation of new lexemes.

The pairs JOUR > DIURNE and SAUVER > SALUT, on the other hand, constitute atypical cases of conversion: according to the definition in Tribout (2010), these pairs would not constitute a morphological relationship, since neither of the two thematic stems formally coincides. Two positions are therefore possible. Either one adheres to Tribout’s strict definition, in which case these two instances do not constitute morphological relations strictly speaking. Or one extends this definition to include suppletive stems, which seems all the more reasonable as it draws a parallel with what is observed in affixation, where cases of allomorphic stems are also accepted.

6.2 Conclusion

This thesis is part of the broader project to develop BREF, which is itself a preliminary step toward the creation of Phononette-3. More specifically, this thesis focuses on the development and evaluation of an intermediate dataset designed to compile the morphological and etymological information necessary for identifying suppletive stems in French.

We first established a methodology for constructing this intermediate dataset using data from Kaikki and relying on LLMs for extraction. Since Kaikki is a structured database of Wiktionary content, it includes an etymological section for each lexeme. From these etymological descriptions, the LLM extracts: the derivational process, the affix, the stem, as well as several additional pieces of information (etymons, source language, semantic base, and etymological derivational family).

To verify the quality of this intermediate dataset, we then developed a two-tier evaluation methodology. First, a human evaluation was conducted in the form of manual annotation based on detailed guidelines for each of the seven questions posed to the LLM. This evaluation provided us with a qualitative overview of the model’s responses and allowed us to identify several recurring patterns (confusion between morphological concepts, difficulty processing non-derived entries, poor handling of the derivational etymological family, etc.). An automated evaluation, calibrated against this manual annotation using heuristic metrics (RMSE, Cohen’s Kappa, Pearson’s r , t -test), was then implemented to assess the quality of the model’s responses across a larger dataset, without having to repeat the manual annotation for each new experiment.

Finally, several avenues for improving the intermediate dataset were explored. First, we extended the etymological context provided to the model by tracing the etymological history of the lexemes back to their base etymon, in order to increase the chances of obtaining a complete description. Second, we compared the performance of several models on this new context. This comparison shows that Qwen, when tested on the extended context, achieves the best results across all models and contexts.

However, the results are not satisfactory enough, and there are many reasons for this. First, the current prompt needs to be revised based on the qualitative observations presented in Chapter 4: in particular, the concepts used (morphological process, affix, stem, base, etymon) should be defined more explicitly within the

¹CONCEPT means ‘chose que l’on peut concevoir’ as well as DANSE means ‘chose que l’on peut danser’ (e.g. “la samba est une danse”).

prompt itself, in order to reduce the confusion observed between these different concepts. Second, the prompt must also be adapted to the model being used: Llama and Deepseek, in particular, did not produce usable responses with the current prompt, suggesting that a single prompt cannot be suitable for all the models tested.

Third, the extension of the etymological context, although motivated by a desire for completeness, had the effect of drowning out relevant information in an excessively large volume of data. It seems necessary to limit or refine this extension.

More generally, this work was only an initial exploratory step, and many additional tests remain to be conducted: expanding the comparison to a larger number of models, more rigorously reevaluating the contribution of few-shot prompting, or exploring other prompting strategies, such as asking questions sequentially (one dedicated prompt per question, while the intermediate dataset is gradually enriched based on the responses obtained) rather than asking all questions at once.

Regarding the ethical aspects of the project, all experiments were conducted on Grid5000. The servers are located in France, a country that relies predominantly on nuclear energy, which limits the carbon footprint of our computations compared to using a service hosted elsewhere. We took care to limit the number of large-scale experiments: processing our 100 reference entries required only between 5 and 15 minutes of computation depending on the models, suggesting a reasonable processing time even at a larger scale (on the order of 8 to 9 hours for 10,000 entries, assuming a linear relationship). This estimate remains to be verified. Furthermore, we systematically validated our approach on a small sample before scaling up, so that we would only need to run the full processing once the methodology was deemed reliable, thereby limiting the use of LLMs.

To summarize my internship, I can say that I encountered several technical challenges. First, working with the models was no easy task. In fact, I faced a huge number of compatibility issues between libraries, which was a real nightmare. For some models, I gave up, and for others, even when I managed to resolve these technical issues, problems persisted. For example, when I used Llama and Deepseek (see Section 5.2): in the end, the prompt was not suited to the model, and the result was not conclusive. Claire GARDENT's courses (*Data Science*, *Generative AI*, and *Prompt Engineering*) helped me a great deal in overcoming these obstacles. Second, I struggled with the code. For a long time, this was a stumbling block in several projects, but thanks to the internship, I feel like I have really made progress in this area. Finally, the linguistics courses (*Morphology* and *Morphophonology*) also contributed significantly to conduct this project.

6.3 Perspectives

The next step will be to apply the methodology developed on a large scale, that is, to all French-derived lexemes available in Kaikki, rather than just to the 100 entries in our reference dataset. This scaling up first requires incorporating the improvements identified in Chapter 4 and 5 (revising the prompt, better defining the expected morphological concepts, and limiting the extended etymological context) to avoid propagating the errors and confusions observed in our sample.

Another approach would be to extend this methodology to languages other than French, provided that the language in question uses suppletive stems in its lexemes formation. We adapted our methodology to Italian by creating a small sample of 23 suppletions from the Italian Kaikki and found that it works just as well in this

language (see in the Appendix D in Table D.3, the comparison of the results between the Italian data and the French data) .

Finally, the most important step will be to transform the resulting intermediate dataset into BREF, that is, into a structured and directly usable database of French suppletive stems. Once BREF is operational, the project to build Phononette can begin, thereby fulfilling the ultimate goal of the larger project by attempting to answer the following question: What is allomorphy and what is suppletion?

Appendix A

Prompts

Prompt	Objectif	Instructions	Few-shot
Simple prompt	Starting point	Short	No
Detailed prompt	To test instructions	Detailed	No
Prompt with few-shot	To test examples	Detailed	Yes
Final prompt	Prompt chose to test on our reference dataset	Detailed	No

TABLE A.1: Prompt summary

This appendix presents the various versions of the prompts developed during this project. They are not independent alternatives, but rather represent the different stages in the process of designing the ideal prompt. Each new version was developed to evaluate the effect of a specific modification on the model’s responses, before arriving at the prompt selected for the final experiments. The main differences (that are in [blue](#)) between these prompts concern the level of detail in the instructions and the use of examples (few-shot prompting). Table [A.1](#) summarizes the main characteristics of each version.

A.1 Prompts for tests

A.1.1 Simple prompt

Tu es un assistant qui répond efficacement aux questions à partir d’un fichier JSON qui contient plusieurs informations à propos d’un mot. Tu ne dois pas donner d’explications, tu ne dois pas raisonner, tu ne dois pas reformuler la tâche.

L’enregistrement en JSON suivant : {json_entry} contient un mot du français, une catégorie, une définition et une étymologie. Construis un nouvel enregistrement JSON qui contient les réponses aux questions suivantes :

- * Q1 = quelle est la langue d’origine du mot qui est indiquée dans l’étymologie ? (un seul mot ou NULL si l’information est absente)
- * Q2 = quelle est la base du mot qui est indiquée dans l’étymologie ? (un seul mot ou NULL si l’information est absente)
- * Q3 = si un préfixe est indiqué dans l’étymologie, quel est ce préfixe ? (un seul mot ou NULL si l’information est absente)
- * Q4 = si un suffixe est indiqué dans l’étymologie, quel est ce suffixe ? (un seul mot ou NULL si l’information est absente)

- * Q5 = si des composants sont indiqués dans l'étymologie, quels sont ces composants ? (liste de composants ou NULL si l'information est absente)
- * Q6 = si le type du procédé morphologique est indiqué dans l'étymologie (suffixation, préfixation, composition, conversion, apocope, etc.), quel est ce type ? (un seul mot ou NULL si l'information est absente)

A.1.2 Detailed prompt

Tu es un assistant d'extraction d'information. Règles : Tu ne dois pas donner d'explications, tu ne dois pas raisonner et tu ne dois pas reformuler la tâche. Tu ne dois rien déduire. Tu dois répondre uniquement avec un JSON valide. Il ne doit y avoir aucun texte avant ou après le JSON. Les réponses sont en minuscules et sans déterminant.

L'enregistrement en JSON suivant : {json_entry} contient un mot du français, une catégorie, une définition et une étymologie. Construis un nouvel enregistrement JSON qui contient les réponses aux questions suivantes :

- * Q1 = quelle est la langue d'origine du mot qui est indiquée dans l'étymologie ? (un seul mot ou NULL si l'information est absente)
- * Q2 = quelle est la base du mot qui est indiquée dans l'étymologie ? (un seul mot ou NULL si l'information est absente)
- * Q3 = si un préfixe est indiqué dans l'étymologie, quel est ce préfixe ? (un seul mot ou NULL si l'information est absente)
- * Q4 = si un suffixe est indiqué dans l'étymologie, quel est ce suffixe ? (un seul mot ou NULL si l'information est absente)
- * Q5 = si des composants sont indiqués dans l'étymologie, quels sont ces composants ? (liste de composants ou NULL si l'information est absente)
- * Q6 = si le type du procédé morphologique est indiqué dans l'étymologie (suffixation, préfixation, composition, conversion, apocope, etc.), quel est ce type ? (un seul mot ou NULL si l'information est absente)

Réponds uniquement avec ce format de JSON :

```
* 'Q1' : ' ',
* 'Q2' : ' ',
* 'Q3' : ' ',
* 'Q4' : ' ',
* 'Q5' : [ ],
* 'Q6' : ' '
```

Si l'information n'est pas explicitement présente dans l'étymologie ou la définition, répondre NULL. Ne jamais déduire ou supposer une information.

A.1.3 Prompt with few-shot

```

Système
---
Tu es un assistant d'extraction d'information.
- Réponds uniquement avec un JSON valide.
- Ne déduis aucune information.
- Si l'information est absente, réponds null.
...

```

Exemple 1

```

-----
Entrée :
{"mot": "ouvrage",
 "catégorie": "Nom commun",
 "définition": "...",
 "étymologie": "Dérivé de ouvrier ... avec le suffixe -age."}

Sortie attendue :
{"Q1": "français",
 "Q2": "ouvrier",
 "Q3": NULL,
 "Q4": "-age",
 "Q5": NULL,
 "Q6": "suffixation"}

```

Exemple 2

```

-----
Entrée :
{"mot": "week",
 "catégorie": "Nom commun",
 "définition": "...",
 "étymologie": "Apocope de week-end."}

Sortie attendue :
{"Q1": NULL,
 "Q2": "week",
 "Q3": NULL,
 "Q4": NULL,
 "Q5": NULL,
 "Q6": "apocope"}

```

A.2 Prompt after tests

The tests conducted in step [A.1.2](#) (detailed prompt) identified several constraints that improve the quality of the responses (requiring the JSON output format, providing explicit rules). These constraints were retained in the final prompt. However, the tests conducted in step [A.1.3](#) (prompt with few-shot) did not reveal any notable improvement in performance (see Section [3.2.3](#)); therefore, this technique was not included in the final prompt as follows:

A.2.1 Final prompt

Tu es un assistant qui répond efficacement aux questions à partir d'un fichier JSON qui contient plusieurs informations à propos d'un mot. Tu ne dois pas donner d'explications, tu ne dois pas raisonner, tu ne dois pas reformuler la tâche.

L'enregistrement en JSON suivant : {json_entry} contient un mot du français, une catégorie, une définition, une étymologie **et une famille de mots**. Construis un nouvel enregistrement JSON qui contient les réponses aux questions suivantes :

- * Q1 = si c'est un mot dérivé, quel est le type de procédé morphologique ? (répondre une liste d'un seul élément s'il n'y a qu'un seul procédé morphologique ou une liste de plusieurs éléments s'il y a plusieurs procédés morphologiques ou une liste vide si le mot n'est pas dérivé)
- * Q2 = quel est l'affixe du mot d'après l'étymologie ? (répondre une liste d'un seul élément s'il n'y a qu'un seul affixe ou une liste de plusieurs éléments s'il y a plusieurs affixes ou une liste vide si le mot n'est pas dérivé)
- * Q3 = quel est le radical du mot ? (répondre null si le mot n'est pas dérivé)
- * Q4 = quels sont les étymons qu'on retrouve dans l'étymologie et quel est leur sens ? (répondre une liste de couples étymon, sens ou une liste vide si le mot n'est pas dérivé)
- * Q5 = quelle est la langue source du mot ? (répondre un seul mot ou null si le mot n'est pas dérivé)
- * Q6 = quelle est la base française du mot ? (répondre un seul mot ou null si le mot n'est pas dérivé)
- * Q7 = à partir de la famille, quels sont les mots ayant le même radical ? (répondre une liste de ces mots ou une liste vide si le mot n'est pas dérivé)

Réponds uniquement avec un unique objet JSON valide de cette forme :

```
{{"Q1": [],
"Q2": [],
"Q3": "",
"Q4": [],
"Q5": "",
"Q6": "",
"Q7": []}}
```

Contraintes strictes :

- Ne renvoie pas une liste
- Ne renvoie pas de texte avant ou après

Appendix B

Script

This is the part of the script that was used to query the LLM Mistral.

```

# import libraries needed
import pandas as pd
import json
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

# set some parameters
shuffle = True # specify that the entries have to be shuffled
max_trials = 5 # max number of trials the LLM has to give an answer
following the expected format
chunk_size = 10 # size of a chunk
device = "cuda" if torch.cuda.is_available() else "cpu" # check if we
have cuda

# load model and tokenizer
MODEL_NAME = "mistralai/Mistral-7B-Instruct-v0.3"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME,
trust_remote_code=True)
model = AutoModelForCausalLM.from_pretrained(MODEL_NAME,
trust_remote_code=True)
model.to(device)
model.eval() # set to eval mode because the model is only used for
inference, not training

all_contexts = pd.read_csv("./context.csv") # all_contexts represents the
csv file of the context automatically generated on Kaikki
if shuffle :
    all_contexts = all_contexts.sample(frac=1, random_state=42) #
randomization of the order of entries
full_df = pd.DataFrame()

for chunk_index in range(0, len(all_contexts) // chunk_size ) :
    chunk = all_contexts.iloc[chunk_index * chunk_size : (chunk_index + 1)
* chunk_size] # we work chunk by chunk
    list_of_dict = chunk.to_dict("records")
    # list_of_dict contains all the entries in the chunk in the form of a
dict
    # these dict will be processed in order to convert them into json

    with open("chunk_context.jsonl", "w", encoding="utf-8") as f :
        for d in list_of_dict :
            f.write(json.dumps(d, ensure_ascii=False) + "\n")

    json_entries = [] # json_entries represents all the entries (and their
context) of the chunk in a jsonl format

    with open("chunk_context.jsonl", "r", encoding="utf-8") as f :
        for line in f :
            json_entry = json.loads(line)
            json_entries.append(json_entry)

    llm_responses = [] # llm_responses represents all the responses
associated to each entry of the chunk

```

```

SYSTEM_PROMPT = """Tu es un assistant qui répond efficacement aux
questions à partir d'un fichier JSON
qui contient plusieurs informations à propos d'un mot. Tu ne dois pas
donner d'explications,
tu ne dois pas raisonner,
tu ne dois pas reformuler la tâche."""

# give 5 chances to the LLM to respond in the right way (the good
format), otherwise, it returns FAIL
nb_trials = 0
json_entry_index = 0
while json_entry_index < len(json_entries) :

    if nb_trials >= max_trials :
        llm_responses.append( {"Q1": ["FAIL"], "Q2": ["FAIL"], "Q3":
"FAIL", "Q4": ["FAIL"], "Q5": "FAIL", "Q6": "FAIL", "Q7": ["FAIL"]} )
        nb_trials = 0
        json_entry_index += 1
        continue

    json_entry = json_entries[json_entry_index]

    prompt = # any prompt we want (one of the prompt presented in
Appendix A)

    messages = [{"role": "system", "content": SYSTEM_PROMPT},
                {"role": "user", "content": prompt}]

    # apply chat template
    formatted_prompt = tokenizer.apply_chat_template(
        messages,
        tokenize=False,
        add_generation_prompt=True
    )

    # tokenize
    inputs = tokenizer(formatted_prompt, return_tensors="pt").to(device)

    # generate
    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=5000, # 1000 is too low
            temperature=0.1,
            do_sample=False,
            pad_token_id=tokenizer.eos_token_id
        )

    # extract the response part by removing the inputs (show response
only)
    generated_tokens = outputs[0][inputs["input_ids"].shape[-1]:]

    # extract the first response [0] since we only input one message

```

```

        content = tokenizer.decode(generated_tokens,
skip_special_tokens=True).strip()
        print("Réponse :", content)

        # data corresponds to the content in a json format
        try :
            data = json.loads(content)
        except Exception as e :
            print("Réponse JSON invalide :", e)
            nb_trials += 1
            continue

        # upper the keys (q1 -> Q1, q2 -> Q2, etc.) to avoid to reject the
response
        new_data = {}
        for k, v in data.items() :
            new_data[k.upper()] = v

        # data is now the response but with uppered keys
        data = new_data

        # check if the format is good (Q1, 2, Q3 way and not "the response of
the first question is: ...")
        if list(data.keys())[0] == "Q1" and list(data.keys())[1] == "Q2" and
list(data.keys())[2] == "Q3" and list(data.keys())[3] == "Q4" and
list(data.keys())[4] == "Q5" and list(data.keys())[5] == "Q6" and
list(data.keys())[6] == "Q7" :
            llm_responses.append(data)
            json_entry_index += 1
            nb_trials = 0
        else :
            nb_trials += 1

        # df_llm_responses are all the valid responses of the llm transformed
in dataframe
        df_llm_responses = pd.DataFrame(llm_responses)
        # name of columns are changed for the question/answer part:
        df_llm_responses = df_llm_responses.rename(columns={"Q1": "proc", "Q2":
"aff", "Q3": "rad", "Q4": "etym_mean", "Q5": "lang", "Q6": "base", "Q7":
"etym_fam"})
        # andfor the context part:
        chunk = chunk.rename(columns={"mot": "lemma", "catégorie": "cat",
"définition": "def", "étymologie": "etym", "famille":
"fam"}).reset_index(drop = True)
        # concatenation of the context part and the question/answer part to be
only one dataframe
        chunk_llm_responses_concat = pd.concat([chunk, df_llm_responses],
axis=1)

        # save the responses of all the chunks when there is no more chunk
        if len(full_df) == 0 :
            full_df = chunk_llm_responses_concat
            full_df.to_csv(f"./Mistral_responses_{'shuffled' if shuffle else
'unshuffled'}.csv", index=False)

```

```
# save chunk by chunk
else :
    full_df = pd.concat([full_df, chunk_llm_responses_concat], axis=0)
    full_df.to_csv(f"./Mistral_responses_{'shuffled' if shuffle else
'unshuffled'}.csv", index=False)
```


Appendix C

Histogramm

Figure C.1 shows the distributions of the human and automatic annotations in the form of a histogram, for the question of the affix. On the x-axis are the three possible scores (0, 0.5, 1) that both the human annotator and the metric can assign to each response, and on the y-axis is the number of responses, ranging from 0 to a maximum of 100 (since our dataset contains exactly 100 entries). The number of responses annotated by the human annotator corresponds to the blue bar, and those annotated by the metric corresponds to the orange one.

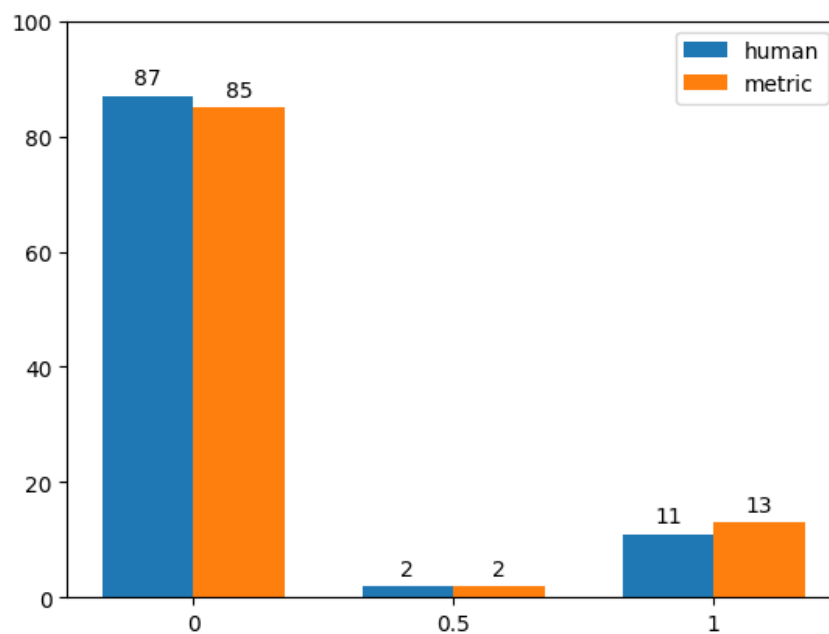


FIGURE C.1: Distribution of scores between the human annotator and the metric, for the question of the affix

Appendix D

Tables

Table D.1 shows the results of the four tests used (RMSE, Cohen’s Kappa, Pearson’s r, t-test) to measure, for each question (proc, aff, rad, etym_mean, lang, base, etym_fam), how closely the scores generated by an automated metric match those given by a human.

Question	RMSE	Cohen’s Kappa	Pearson’s r	t-test
proc	0.07	0.96	0.98	0.16
aff	0.10	0.84	0.96	0.04
rad	0.09	0.93	0.98	0.57
etym_mean	0.20	0.77	0.89	1.00
lang	0.00	1.00	1.00	NaN
base	0.09	0.90	0.97	0.57
etym_fam	0.09	0.69	0.89	0.08

TABLE D.1: Measurement and test results for every question

Table D.2 shows the results obtained by each model (Mistral, Qwen), based on a specific context (initial context or extended context) and by question type (proc, aff, rad, etym_mean, lang, base, etym_fam). For each question, the average of all scores assigned by the metric was calculated. Therefore, if the average score for a question is 1, this means that our metric assigned only 1s and that the LLM answered perfectly for all entries.

	Mistral (initial context)	Mistral (extended context)	Qwen (extended context)
proc	0.71	0.57	0.57
aff	0.14	0.10	0.29
rad	0.24	0.16	0.45
etym_mean	0.33	0.13	0.17
lang	0.45	0.41	0.44
base	0.16	0.08	0.05
etym_fam	0.03	0.01	0.10
average	0.29	0.21	0.30

TABLE D.2: Comparison of the results between the initial context and the extended context with different models

Table D.3 shows the results of each model on the suppletive data only. The first column shows the results of the Mistral model on the initial context; the second column shows the results of the Mistral model on the extended context; the third column shows the results of the Qwen model on the extended context; and the fourth column shows the results of the Qwen model on Italian, with a context that has not been extended. For each question, the average of all scores was calculated.

	Mistral (initial context)	Mistral (extended context)	Qwen (extended context)	Qwen (italian)
proc	0.49	0.47	0.40	0.26
aff	0.26	0.19	0.45	0.48
rad	0.32	0.18	0.45	0.37
etym_mean	0.25	0.18	0.23	0.24
lang	0.86	0.80	0.84	0.83
base	0.26	0.12	0.04	0.00
etym_fam	0.04	0.03	0.15	0.48
average	0.35	0.28	0.37	0.38

TABLE D.3: Comparison of the results between the initial context, the extended and between French and Italian, on the suppletive data only

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